

LATE-QUATERNARY VEGETATION DYNAMICS IN NORTH AMERICA: SCALING FROM TAXA TO BIOMES

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Abstract. This paper integrates recent efforts to map the distribution of biomes for the late Quaternary with the detailed evidence that plant species have responded individually to climate change at millennial timescales. Using a fossil-pollen data set of over 700 sites, we review late-Quaternary vegetation history in northern and eastern North America across levels of ecological organization from individual taxa to biomes, and apply the insights gained from this review to critically examine the biome maps generated from the pollen data. Higher-order features of the vegetation (e.g., plant associations, physiognomy) emerge from individualistic responses of plant taxa to climate change, and different representations of vegetation history reveal different aspects of vegetation dynamics. Vegetation distribution and composition were relatively stable during full-glacial times (21 000–17 000 yr BP) [calendar years] and during the mid- to late Holocene (7000–500 yr BP), but changed rapidly during the late-glacial period and early Holocene (16 000–8 000 yr BP) and after 500 yr BP. Shifts in plant taxon distributions were characterized by individualistic changes in population abundances and ranges and included large east–west shifts in distribution in addition to the northward redistribution of most taxa. Modern associations such as *Fagus–Tsuga* and *Picea–Alnus–Betula* date to the early Holocene, whereas other associations common to the late-glacial period (e.g., *Picea–Cyperaceae–Fraxinus–Ostrya/Carpinus*) no longer exist. Biomes are dynamic entities that have changed in distribution, composition, and structure over time. The late-Pleistocene suite of biomes is distinct from those that grew during the Holocene. The pollen-based biome reconstructions are able to capture the major features of late-Quaternary vegetation but downplay the magnitude and variety of vegetational responses to climate change by (1) limiting apparent land-cover change to ecotones, (2) masking internal variations in biome composition, and (3) obscuring the range shifts and changes in abundance among individual taxa. The compositional and structural differences between full-glacial and recent biomes of the same type are similar to or greater than the spatial heterogeneity in the composition and structure of present-day biomes. This spatial and temporal heterogeneity allows biome maps to accommodate individualistic behavior among species but masks climatically important variations in taxonomic composition as well as structural differences between modern biomes and their ancient counterparts.

Key words: biome maps; biome reconstruction, pollen-based; fossil pollen data; North America, vegetation history; paleoecology; plant functional types; Quaternary; vegetation dynamics; vegetation history.

INTRODUCTION

Vegetational responses to late-Quaternary environmental change were large, complex, and spanned a wide range of spatial, temporal, and ecological scales. Mapped syntheses of fossil pollen data enable spatially explicit studies of vegetational history (Huntley and

Birks 1983, Thompson 1988, Webb 1988, PALE Beringian Working Group 1999) across ecological levels ranging from individual species to biomes. Previous syntheses have demonstrated that, at subcontinental spatial scales and millennial timescales, plant taxa responded individually to late-Quaternary environmental change (Davis 1981b, Huntley and Birks 1983, Huntley and Webb 1988, Webb 1988). Plant associations (Gleason 1926) emerge from the individualistic behavior of plant taxa (Jacobson et al. 1987), and plant associations existed in the past that have no floristic counterpart today (Cushing 1967, Overpeck et al. 1992, Williams et al. 2001). Changes in the distribution of individual plant taxa also scale up to alter vegetation physiognomy at continental to global scales (mapped

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as the distribution of plant functional types and biomes) and thereby affect the exchanges of energy, moisture, aerosols, and trace gases between the land surface and atmosphere (Foley 1994, Harrison et al. 1995, Kutzbach et al. 1996, Pielke et al. 1998). Understanding the role of vegetation as an active component of the Earth system therefore requires study across a broad range of temporal, spatial, and ecological scales.

Global biome distributions have been reconstructed for the mid-Holocene and last glacial maximum from fossil pollen and macrofossil records (Jolly et al. 1998, Prentice and Webb 1998, Williams et al. 1998, 2000, Elenga et al. 2000, Prentice et al. 2000, Takahara et al. 2000, Tarasov et al. 2000, Thompson and Anderson 2000, Yu et al. 2000, Gotanda et al. 2002). The resultant vegetation maps are at an ecological level of organization suitable for interfacing with Earth-system models. The key conceptual advance has been to define biomes as assemblages of plant functional types (PFTs), rather than in floristic terms, enabling pollen-based biomes to be mapped globally (Prentice et al. 1996, Prentice and Webb 1998). Definition of plant function is scale dependent (Smith et al. 1997); globally scaled definitions of PFTs typically are based on life-form, phenology, leaf shape, and climatic tolerances (Cramer 1997, Prentice and Webb 1998). The inferred biome maps can provide realistic prescribed vegetation maps for general-circulation model experiments or a standard of comparison for evaluating global vegetation model simulations (Texier et al. 1997, Broström et al. 1998, Jolly et al. 1998, Prentice et al. 1998, Williams et al. 1998, Joussaume et al. 1999). The Biome 6000 consortium (Prentice and Webb 1998, Kohfeld and Harrison 2000) adopted the Prentice biomization method as a common technique for assigning fossil pollen and macrofossil assemblages to PFTs and biomes (Prentice et al. 1996, Prentice and Webb 1998). The biomization method has also been applied to individual time series to reconstruct long-term fluctuations in vegetation (Allen et al. 1999, Gotanda et al. 2002, Marchant et al. 2002).

In scaling up from multivariate pollen spectra to biomes, ecological information is inevitably lost, and the biome maps require critical examination. The assignments of the biomization method, like those of any other vegetation classification scheme, are subject to error. More broadly, as categorical representations of the vegetation, the biome maps imply a temporal and spatial homogeneity, which ignores the continuous nature of most vegetation gradients (Whittaker 1956, DeFries et al. 1999, Colinvaux et al. 2000) and masks temporal variations in these gradients. The division of the past vegetation into a discrete number of biomes seems inconsistent with the evidence that plants responded individually to late-Quaternary climate change (Davis 1981b, Huntley and Birks 1983, Huntley and Webb 1988, Webb 1988).

This paper reviews the major features of late-Quaternary vegetation history in northern and eastern North America and critically evaluates the biome interpretations generated from the pollen data. We show how the individualistic shifts in range and abundance for plant taxa scale upward to cause (1) compositional shifts within plant communities, (2) appearances and disappearances of novel plant associations, and (3) changes in the position, area, composition, and structure of biomes. These events are illustrated by mapped and time-series representations of vegetation history based upon fossil pollen data from boreal and eastern North America for the past 21 000 calendar years (21 000 BP) in which we shift in focus from lower-order (higher resolution) ecological levels of organization (individual taxa, plant associations) to higher-order (lower resolution) features of the vegetation (aggregate rates of change, biomes). Each visualization highlights different aspects of vegetation history at various ecological resolutions, and together they provide a) a comprehensive view of vegetation dynamics in North America during the past 21 000 yr and b) a basis for assessing the biome maps. Understanding the qualities of biome-based reconstructions compared to other representations of vegetation change is essential to their informed use with Earth-system models.

DATA AND METHODS

The pollen data used in this study are drawn primarily from the North American Pollen Database (NAPD),⁶ supplemented by data for the mid-Holocene and last glacial maximum from the Base de Données Polliniques et Macrofossiles du Québec (BDPMQ; P. Richard, personal communication), the Paleoclimates from Arctic Lakes and Estuarines Project (PALE; M. Duvall, personal communication), and the digitization of published pollen diagrams. Many of the sites used in this study have been included in previous syntheses of eastern (Webb 1987, 1988, 1993, Jackson et al. 2000, Williams et al. 2000) and western (Ritchie 1976, Anderson et al. 1989, Anderson and Brubaker 1993, Ritchie and Harrison 1993, Williams et al. 2000) North American pollen records. A full description of the pollen data set is available elsewhere (Williams 2000, Shuman 2001). In total, we assembled 759 fossil-pollen sites, with the number of sites per time interval steadily increasing over time (Fig. 1), a trend largely caused by the progressive increase in habitable land area as the Cordilleran and Laurentide ice sheets retreated. Data from the western United States, with the exception of the Pacific Northwest, was excluded from this study because of the complexity of the regional topography and the low density of pollen sites available in the NAPD. Syntheses of paleoecological data from the western United States are available elsewhere (Thomp-

⁶ URL: (<http://www.ngdc.noaa.gov/paleo/napd.html>)

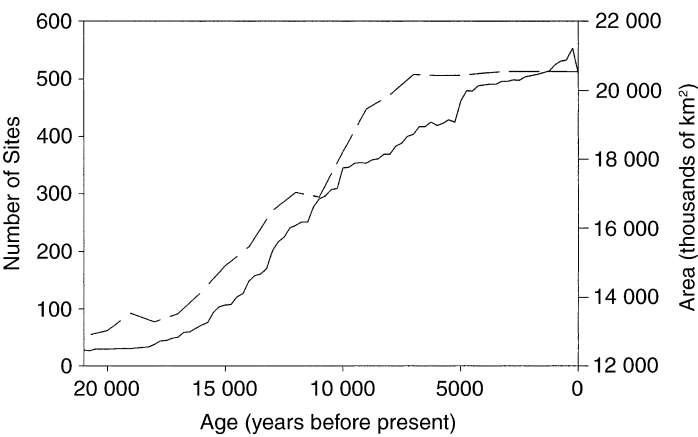


FIG. 1. The number of pollen sites available in boreal and eastern North America per time interval (solid line) and the total unglaciated land area in North America (dashed line).

son 1988, Thompson et al. 1993, Thompson and Anderson 2000).

We compiled pollen percentages for 61 taxa common in North American pollen records (Table 1) using the same taxa for the pollen sum. To calculate pollen-sample ages, we linearly interpolated between the calibrated ages of age-control points in the preferred age models provided by the NAPD, BDPMQ, PALE, and their contributors, except as noted in Shuman (2001). Radiocarbon dates were converted to calendar years by linear interpolation between radiocarbon dates in the INTCAL98 data set (Stuiver et al. 1998). In cases where a radiocarbon age had multiple intercepts on the calibration curve, we chose the median calendar age. We linearly interpolated pollen percentages to each 1000-year interval from the nearest older and younger pollen samples. We also included the 500 yr BP time window to represent vegetation at the point prior to European settlement.

To ensure that we did not over-interpolate between widely spaced pollen samples and/or age controls, we

ranked the quality of the age constraints and pollen sampling for each 1000-yr interval. Acceptable age controls for the pollen records included radiocarbon dates (conventional and AMS), ash/tephra layers, the *Tsuga canadensis* decline around 5400 yr BP (Webb 1982, Allison et al. 1986), the European settlement horizon, and the assignment of 0 yr BP to core tops. For each pair of bracketing dates, we define an “age envelope” in which the older age is equal to the age of the older control plus its 1-σ uncertainty and the younger age is equal to the age of the younger control minus its 1-σ uncertainty. The age rank *R* for each interval is equal to the minimum age difference between the age of the interval (*A_i*) and the two envelope boundaries. An analogous procedure was used to assess the pollen sampling quality for each time interval, after using the age model to assign ages to each pollen sample. The information about the quality of age controls (*R_a*) and pollen sampling (*R_p*) at each site was used to exclude sites from mapping and to weight sites in the spatial interpolation (see below).

TABLE 1. The 61 taxa in the pollen data set, categorized by life-form.

Life-form	Taxa
Needle-leaved woody	<i>Abies</i> (fir), Cupressaceae/Taxaceae undif. (Cypress/Yew families), <i>Larix/Pseudotsuga</i> (larch/Douglas fir), <i>Picea glauca</i> (white spruce), <i>Picea mariana</i> (black spruce), <i>Picea</i> undif. (spruce), <i>Pinus strobus</i> (eastern white pine), <i>Pinus</i> undif. (pine), <i>Taxodium</i> (cypress), <i>Thuja</i> (red cedar), <i>Tsuga</i> (hemlock)
Broad-leaved woody	<i>Acer</i> (maple), <i>Alnus</i> (alder), Aquifoliaceae (Holly family), <i>Betula</i> (birch), <i>Carya</i> (hickory), <i>Castanea</i> (chestnut), <i>Ceanothus</i> (ceanothus), <i>Celtis</i> (hackberry), <i>Cephalanthus</i> (button-bush), <i>Cercocarpus</i> (mountain mahogany), <i>Clethra</i> (sweetpepperbush), <i>Corylus</i> (hazel), <i>Empetrum</i> (crowberry), Ericaceae (Heath family), <i>Fagus</i> (beech), <i>Fraxinus</i> (ash), <i>Juglans</i> (walnut), <i>Liquidambar</i> (sweetgum), <i>Mimosa</i> (mimosa), Myricaceae (Myrica family), <i>Nyssa</i> (tupelo), <i>Ostrya/Carpinus</i> (hornbeam/hophornbeam), <i>Platanus</i> (sycamore), <i>Populus</i> (poplar), <i>Quercus</i> (oak), <i>Salix</i> (willow), <i>Shepherdia</i> (buffaloberry), <i>Tilia</i> (Basswood), <i>Ulmus</i> (elm)
Herb/Small shrub	<i>Ambrosia</i> (ragweed), Apiaceae (Parsley family), <i>Artemisia</i> (sagebrush), Asteraceae undif. (Aster family), Brassicaceae (Mustard family), Caryophyllaceae (Pink family), Chenopodiaceae/Amaranthaceae (Goosefoot/Pigweed families), Cyperaceae (Sedge family), <i>Dryas</i> (mountain-avens), <i>Ephedra</i> (jointfir), <i>Eriogonum</i> (buckwheat), Euphorbiaceae (Spurge family), Fabaceae (Pea family), <i>Oxyria</i> (mountainsorrel), Poaceae (Grass family), Ranunculaceae (Buttercup family), Rubiaceae (Madder family), <i>Sarcobatus</i> (greasewood), Saxifragaceae (Saxifrage family), <i>Sphaeralcea</i> (globemallow)
Moss	<i>Selaginella</i> (spikemoss)

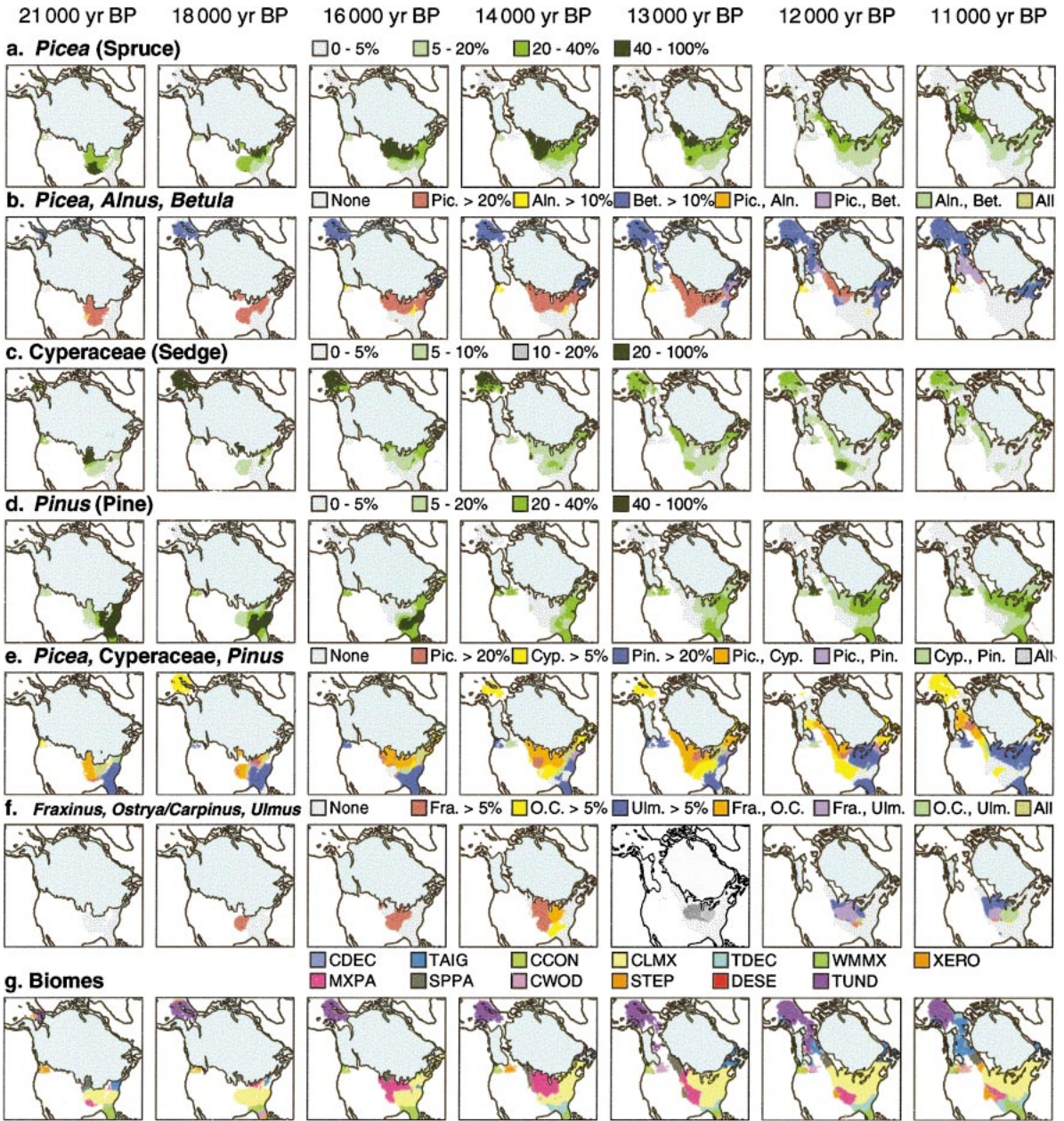


FIG. 2. Single-taxon isopoll maps, group isopoll maps, and inferred biome distributions in boreal and eastern North America since the last glacial maximum. (a, c, d) In the single-taxon maps, the pollen abundances of a single taxon are displayed as various shades of green, with high color saturations corresponding to high abundances. Regions with insufficient data for mapping are left blank. (b, e, f) In the multi-taxon isopoll maps, each of three pollen taxa is mapped as “present” or “absent,” and the eight possible combinations of presence and absence are each mapped as a distinct color (Jacobson et al. 1987). Primary colors (red, blue, cyan) indicate regions where only one taxon is present in abundance. Secondary colors (orange, purple, and green) indicate associations between pairs of taxa; areas where all three taxa are associated are beige. The abundance threshold chosen for each pollen type is set relatively high to indicate only those regions where the taxon was an important constituent of the regional vegetation (Jacobson et al. 1987). (g) Biome distribution. Key: CCON = cool conifer forest, CDEC = cold deciduous forest, CLMX = cool mixed forest, CWOD = conifer woodland, DESE = desert, MXPA = mixed parkland, SPPE = spruce parkland, STEP = steppe, TAIG = taiga, TDEC = temperate deciduous forest, TUND = tundra, WMMX = warm mixed forest, XERO = xerophytic scrub. By reading the maps horizontally, one can track the history of a single plant taxon, plant association, or biome. Vertical comparisons across maps provide information about the interplay among ecological resolutions. Animated versions of these maps and others not shown here may be viewed as a supplement or at the NOAA web site (see footnote 7). For all maps, map projection is Albers equal area with standard parallels 33.33° N and 66.66° N, center point = 70° N, 100° W.

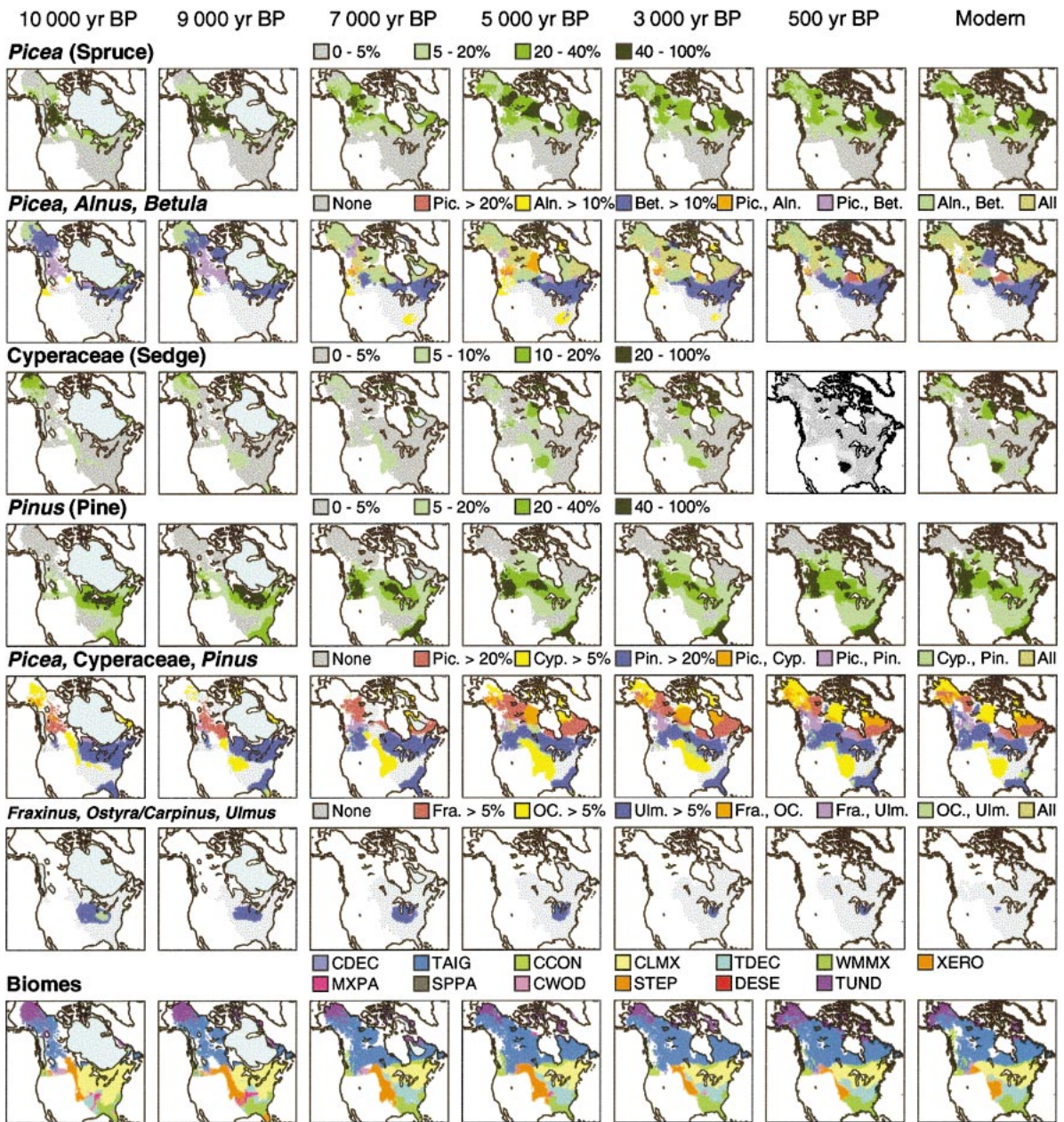


FIG. 2. Continued.

Spatial interpolation and paleogeography

To present a continuous coverage of vegetation patterns, the fossil-pollen abundances for each interval were spatially interpolated from individual sites to a 50-km grid. A search window measuring 300 × 300 km (horizontal) × 500 m (vertical) was extended from each gridpoint, and pollen percentages from all sites within the window were interpolated to the grid center by averaging pollen percentages within the window according to a tri-cubic distance weighting (Huntley et al. 1989, Cleveland 1993, Webb et al. 1993). Pollen sites were not used for an interval if $R_a > 5000$ years

or if $R_p > 2000$ years. The distance was calculated across three spatial dimensions, representing the distance between the gridpoint and pollen site, and two temporal dimensions, representing the distance between the interpolated time interval and the nearest pollen sample and age control at the site (R_a and R_p). Choosing a large window size smoothes out site-specific variability in the pollen data, which may reflect edaphic controls or local responses to disturbances (Graumlich and Davis 1993), and highlights the regional patterns in the vegetation. Including R_a and R_p in distance-weighting downweights sites with poor age

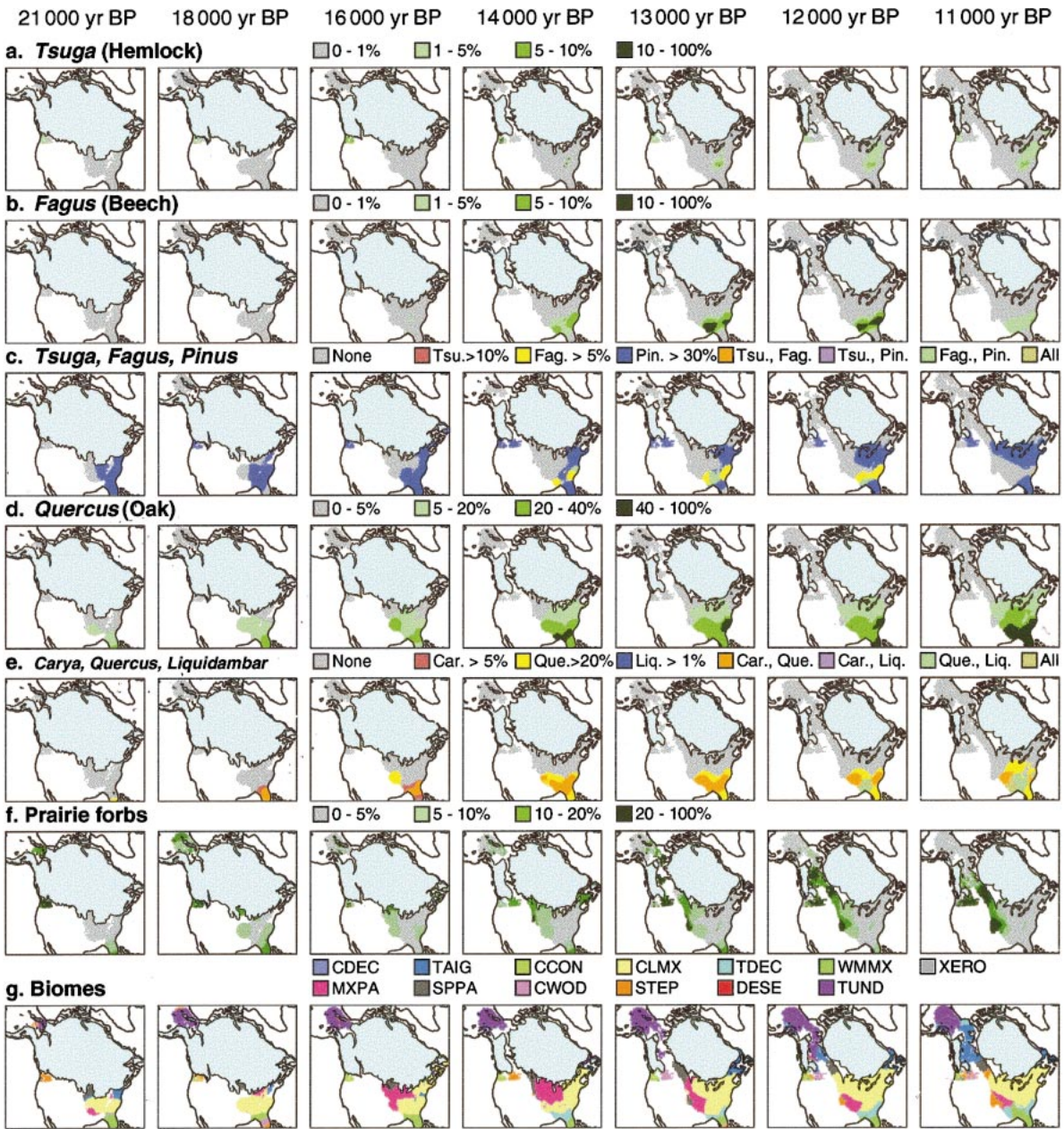


FIG. 3. Single-taxon isopoll maps, group isopoll maps, and inferred biome distributions (repeated from Fig. 2 for comparison) in boreal and eastern North America since the last glacial maximum. The format is as in Fig. 2; maps are for *Tsuga*, *Fagus*, *Tsuga*–*Fagus*–*Pinus*, *Quercus*, *Carya*–*Quercus*–*Liquidambar*, prairie forbs, and biomes. The prairie forb category comprises Asteraceae and Chenopodiaceae/Amaranthaceae.

controls and/or pollen sampling. Grid points were left empty if no sites were within the search window or if the combined tri-cubic weight of all sites within the window was less than 0.5, a threshold chosen by experimentation (Williams 2000).

The past positions of the Laurentide and Cordilleran Ice Sheets were digitized from Dyke and Prest (1987) and the National Atlas of Canada (Canada Geographical Services Division 1985). These maps were ad-

justed according to radiocarbon calibration (Stuiver et al. 1998) and in accordance with the findings of Barber et al. (1999). The past position of North American coastlines was determined by interpolating the Peltier (1994) topographic anomalies (differences in elevation between, e.g., 21 000 yr BP and the present) onto a 5-min digital elevation model, and then determining the location of the 0-m contour. The position of Lake Agassiz was derived from Teller et al. (1983).

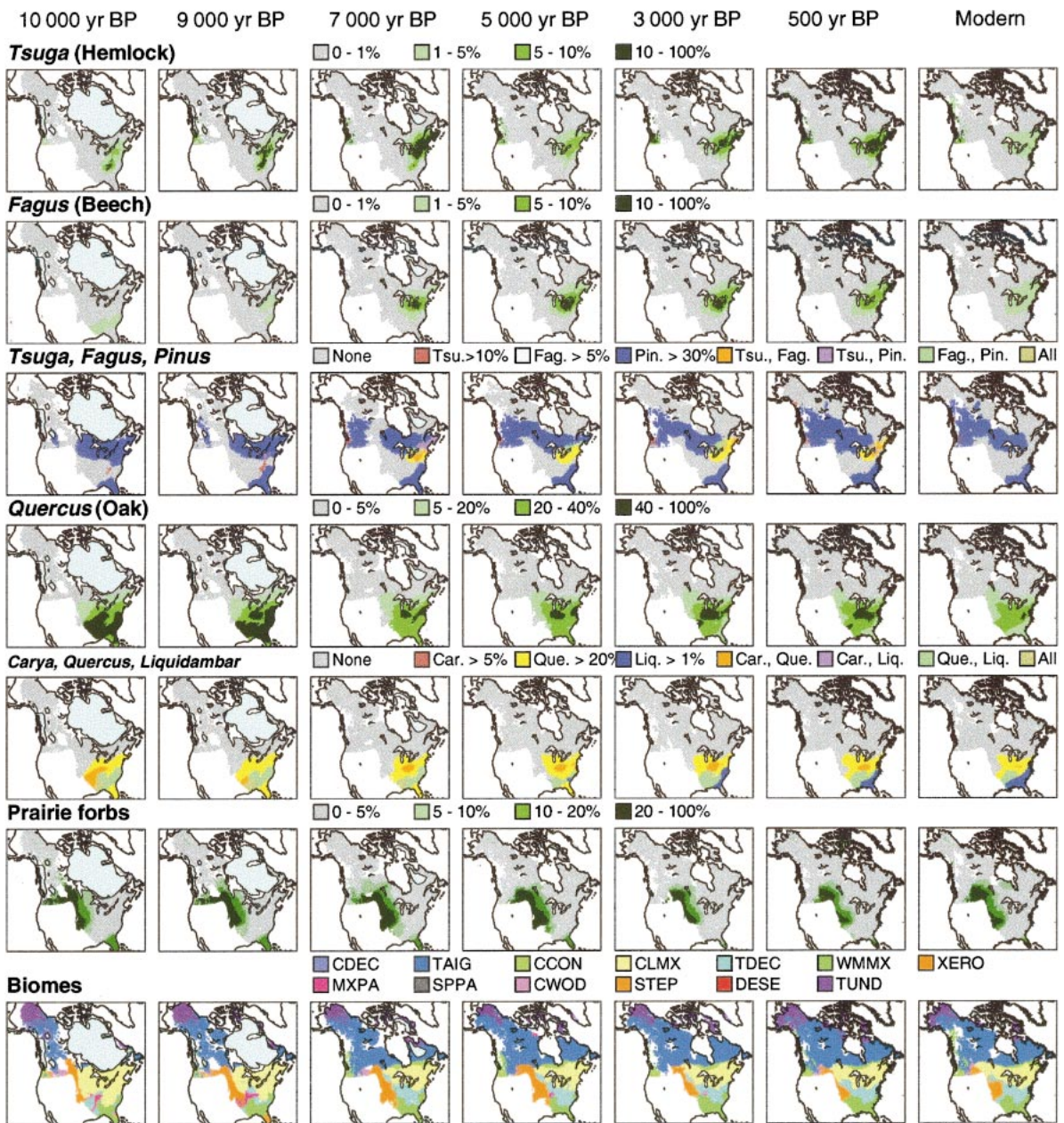


FIG. 3. Continued.

Data visualizations

We present five visualizations of the pollen data: isopoll maps for a single taxon, isopoll maps for groups of taxa, dissimilarity maps showing rates of vegetation change, maps of biomes, and time-series plots of biome composition. We interleave the different kinds of maps to facilitate cross comparison (Figs. 2–4) but describe each map series or time series separately. Mapping all pollen taxa and combinations for all 1000-yr intervals is impractical, so we present here a representative selection of maps and intervals and refer the reader to the Supplement or to the U.S. National Oceanic and

Atmospheric Administration (NOAA) web site for supplementary maps and animations.⁷

Isopoll maps.—Isopoll maps show the distributions of individual pollen types at regional to subcontinental scales (Davis and Webb 1975, Webb and McAndrews 1976, Bernabo and Webb 1977, Webb et al. 1983b, Delcourt and Delcourt 1987, Jacobson et al. 1987). The present-day distribution of pollen percentages is spatially similar to mapped patterns of plant dominance and density (Delcourt et al. 1984, Bradshaw and Webb

⁷ URL: <www.ngdc.noaa.gov/paleo/pubs/williams2004>

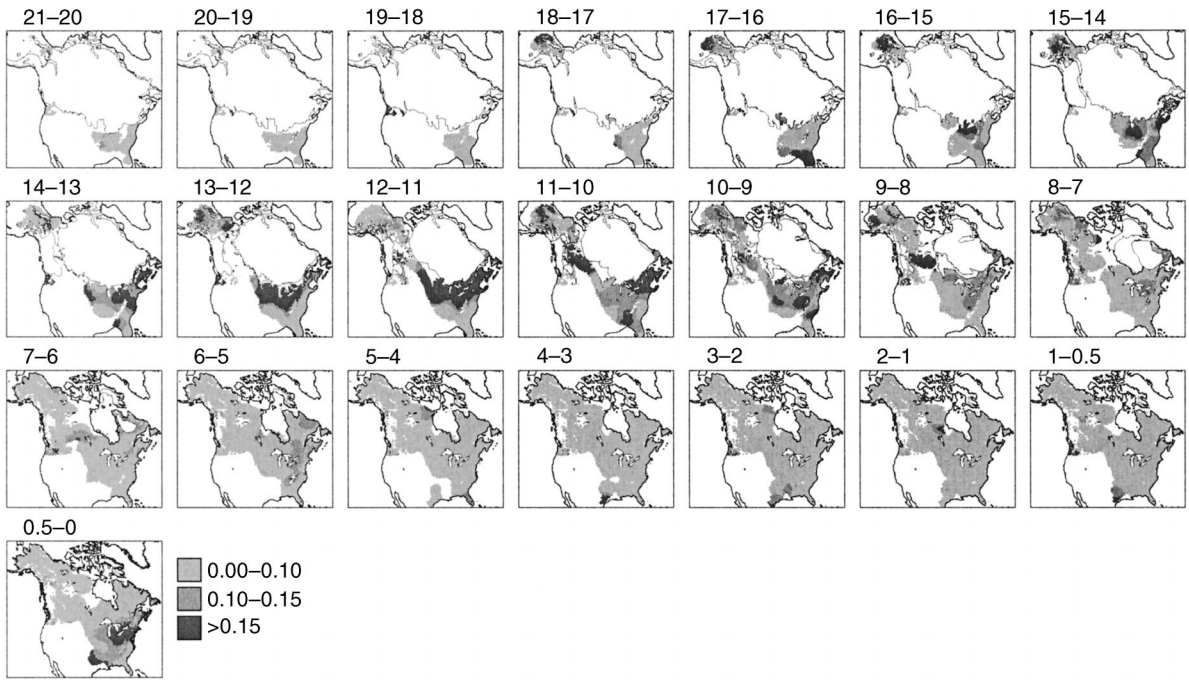


FIG. 4. Maps of the squared-chord dissimilarity (SCD) between adjacent time intervals. To measure vegetation change, dissimilarities were calculated only within core (Overpeck et al. 1991, Grimm and Jacobson 1992). Because all intervals are equally spaced in time (except for 500 yr BP), the mapped dissimilarity values represent both the magnitude and rate of vegetation change. For comparison, modern pollen samples drawn from different vegetation formations typically have SCDs > 0.15 (Overpeck et al. 1985).

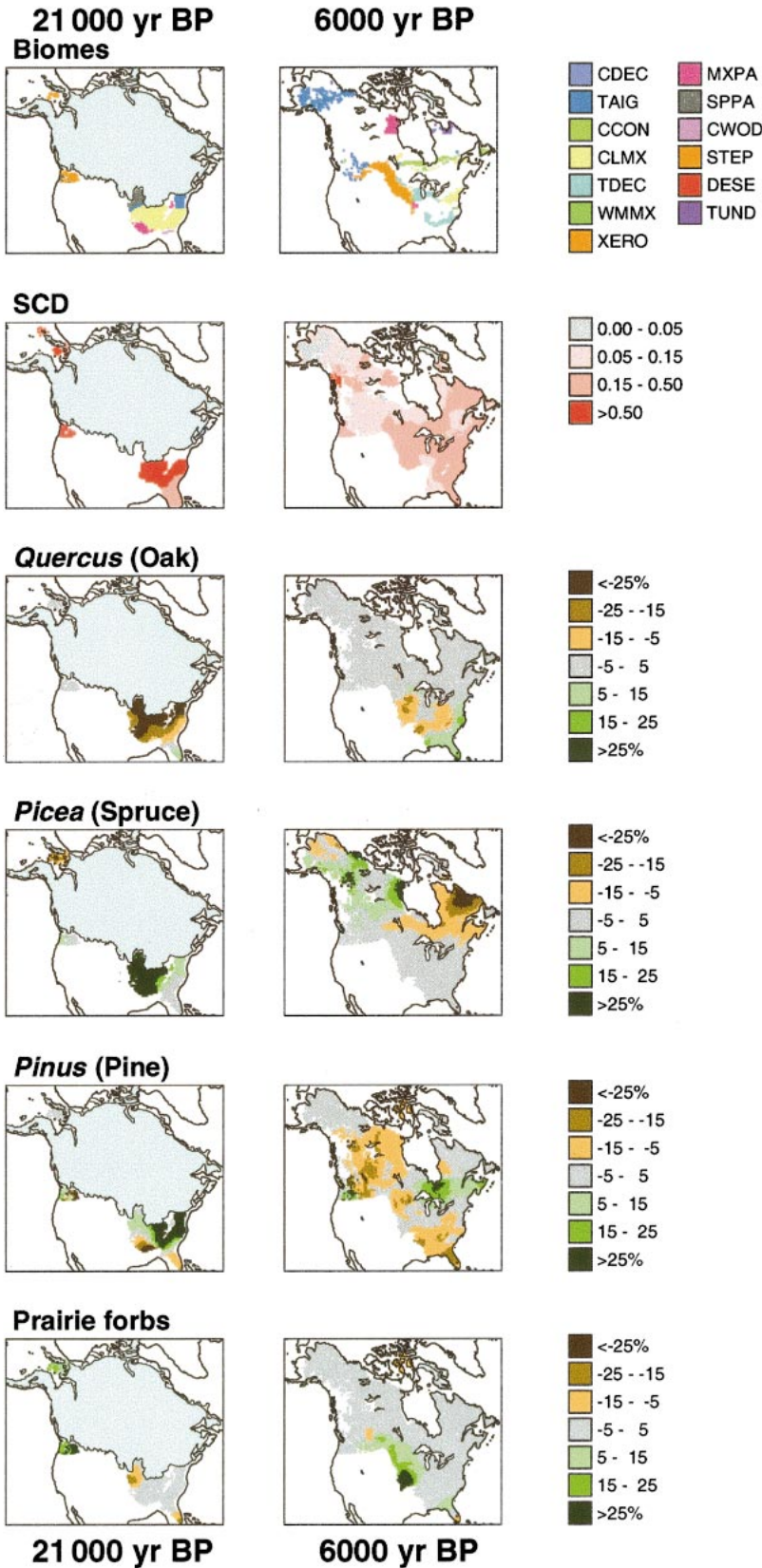
1985, Solomon and Webb 1985), although direct calibration is hampered by intertaxonomic differences in pollen source area and productivity (Prentice 1985, 1988, Sugita 1994, Sugita et al. 1999). We map the interpolated pollen percentages for each grid point as various shades of green (Figs. 2 and 3), producing maps equivalent to traditional isopoll maps (Bernabo and Webb 1977). The scaling of the isopoll contours for each pollen type was based upon its abundance in the fossil record (Webb 1988, Webb et al. 1993). We assume that pollen percentages correspond positively to plant densities (Delcourt et al. 1984, Bradshaw and Webb 1985). The smallest contour provides a rough index of plant range, but is only an approximation—it misses outlier populations not sampled by the pollen record (Kullman 1995) or represented by pollen percentages less than the contour (Bennett 1985). Animated versions of these mapped time series have been archived at the World Data Center for Paleoclimatology

(see footnote 7). Mapped series referred to in the text but not shown here may be viewed on-line, at the U.S. NOAA web site (see footnote 7).

Multi-taxon isopoll maps (Jacobson et al. 1987) are ideal for showing how the unique trajectories of individual taxa scale up to form plant associations. Plant associations are defined in floristic terms and are represented in the multitaxon isopoll maps by the overlaps among isopolls for different taxa. Each taxon is represented by a single percentage level, and each of eight possible taxon combinations is represented as a unique color. The choice of percentage levels is intended to represent the point at which the taxon is a dominant or subdominant component of the vegetation or illustrates clearly an interaction with other taxa or taxon combinations. The appearance or disappearance of colors signifies changes in plant associations.

Rates of change.—Past rates of vegetational change are inferred from squared chord distances (Overpeck et

FIG. 5. Anomaly maps for biomes, squared-chord distances (SCD), *Quercus*, *Picea*, *Pinus*, and prairie forbs for the last glacial maximum vs. pre-settlement vegetation (21 000 yr BP vs. 500 yr BP), and mid-Holocene vs. pre-settlement (6000 yr BP vs. 500 yr BP). The biome anomaly maps show the past biome assignments for the grid points that differ in biome type between past and pre-settlement vegetation. Areas with no data or with unchanged biome assignments are left blank. The dissimilarity maps show the aggregate palynological differences between the past and present; darker reds indicate higher dissimilarities. In the anomaly maps for individual plant taxa, darker browns indicate that the taxon was less abundant in the past; green indicates that the taxon was more abundant in the past. For biome key see Fig. 2 legend.



al. 1985) between adjacent 1000-yr intervals at each location (Jacobson et al. 1987, Overpeck et al. 1991). The squared chord distances were calculated for the gridded pollen data set rather than the original site data. Squared-chord distances larger than 0.15 suggest a change in vegetation composition equivalent in scale to the differences among modern vegetation formations (Overpeck et al. 1985).

Biomes.—The gridded fossil-pollen data are classified into biomes via the biomization method (Prentice et al. 1996, Prentice and Webb 1998), as adapted for use with North American pollen taxa (Williams et al. 1998, 2000, Thompson and Anderson 2000). In the biomization method, biome types are defined a priori as assemblages of plant functional types, permitting the recognition of biomes not found in the modern vegetation. By introducing plant functional types (PFTs) as an intermediate stage between individual pollen taxa and biomes, biomes are defined as physiognomic entities potentially global in scope, with different floristic representatives among regions. The biomization method is most sensitive to the presence or absence of plant functional types, and less sensitive to their relative proportions. Therefore, both taxonomic composition and the abundances of PFTs may vary within biomes. The biome definitions used in this paper are consistent with those for other regions of the world (Prentice et al. 2000), but we have expanded the list of biomes to include a mixed parkland and spruce parkland, biomes that apparently were common in the late Pleistocene but are rare or absent today (Williams et al. 2001).

Anomaly maps.—Data–model comparisons frequently employ “anomaly maps” (differences between the past and present distributions of the variable of interest) in order to minimize the effects of model bias (e.g., Harrison et al. 1995, Webb et al. 1998) and to visually highlight differences between the present and past. To illustrate how different visualizations affect the mapped patterns of anomalies, we chose two time slices—21 000 yr BP and 6000 yr BP—roughly equivalent to the time intervals (18 000 and 6000 radiocarbon year BP) mapped by Biome 6000 (Prentice and Webb 1998), and differenced from 500 yr BP in order to isolate the effects of late-Quaternary environmental change from European land use.

Biome composition plots.—When producing the composition plots, all samples assigned to each biome for each 1000-yr interval were grouped and the pollen percentages averaged. In order to compare spatial and temporal variability in biome composition, time-series plots of the mean pollen abundances are juxtaposed with box plots displaying the variation of pollen percentages for present-day biomes. Plant taxa were additionally assigned to plant life-forms (Table 1) and pollen percentages summed to estimate the relative proportions of plant life-forms for each biome.

LATE-QUATERNARY VEGETATION HISTORY

Individual taxa

The single-taxon isopoll maps and animations (Figs. 2a, c, and d, and 3a, b, d, and f) illustrate the fluidity of plant distributions at continental and millennial scales. All plant taxa have experienced major changes in range and abundance since the last glacial maximum.

Plant distribution shifts have spanned large sections of North America and included both north-south and east-west components.—Arboreal taxa expanded north after the last glacial maximum in response to increasing temperatures and retreat of northern hemisphere ice sheets. Northern populations of boreal taxa (e.g., *Picea*, *Pinus*, *Abies*, *Larix*, and *Betula*) expanded in areas vacated by the retreating ice sheets while southern populations declined. Temperate trees increased in abundance and their distributions shifted north from the southeastern United States. *Quercus* abundances, for example, were low and apparently limited to the southeastern United States at 21 000 yr BP, but its range (approximated by the 5% isopoll) expanded rapidly northward between 21 000 yr BP and 14 000 yr BP (Fig. 3d). The patterns of population expansion differ among taxa: several taxa (*Tsuga*, *Castanea*) appear to have been most abundant in the Appalachians while others (*Quercus*, *Carya*) appear to have been most abundant at lower elevations (Fig. 3a,b; see also the NOAA web site [footnote 7]).

Within the context of northward migration, east–west shifts in plant distributions also occurred, particularly for moisture-sensitive plant taxa and boreal taxa occupying areas vacated by the Laurentide Ice Sheet. Between 13 000 and 11 000 yr BP, *Pinus* pollen abundances rapidly increased in the Great Lakes region as *Picea* and Cyperaceae pollen abundances decreased (Fig. 2a,c,d, and e). The timing of the pine increase at some sites in Minnesota and Wisconsin may shift to younger ages as new accelerator mass-spectrometry (AMS) dates replace bulk radiocarbon chronologies (E. C. Grimm and L. J. Maher, *unpublished manuscript*), but the broad mapped patterns should persist. Regionally lowered lake-levels coincident with the 13 000–11 000 yr BP *Pinus* expansion indicate that *Pinus* was favored in part by a regional drop in moisture availability (Shuman et al. 2002b). Prairie forb abundances increased during the early Holocene then declined in response to fluctuations in moisture balance (McAndrews 1966, Wright 1968, Webb et al. 1983a, Winkler et al. 1986, Baker et al. 2002). The timing of these variations, however, differed regionally (Baker et al. 1992).

These meridional shifts in taxon distributions span much of eastern and western North America, particularly in the higher latitudes. Both *Picea* and *Pinus* became more widely abundant after the last glacial maximum, by increasing in abundance north and west into central Canada, thus linking with western species ex-

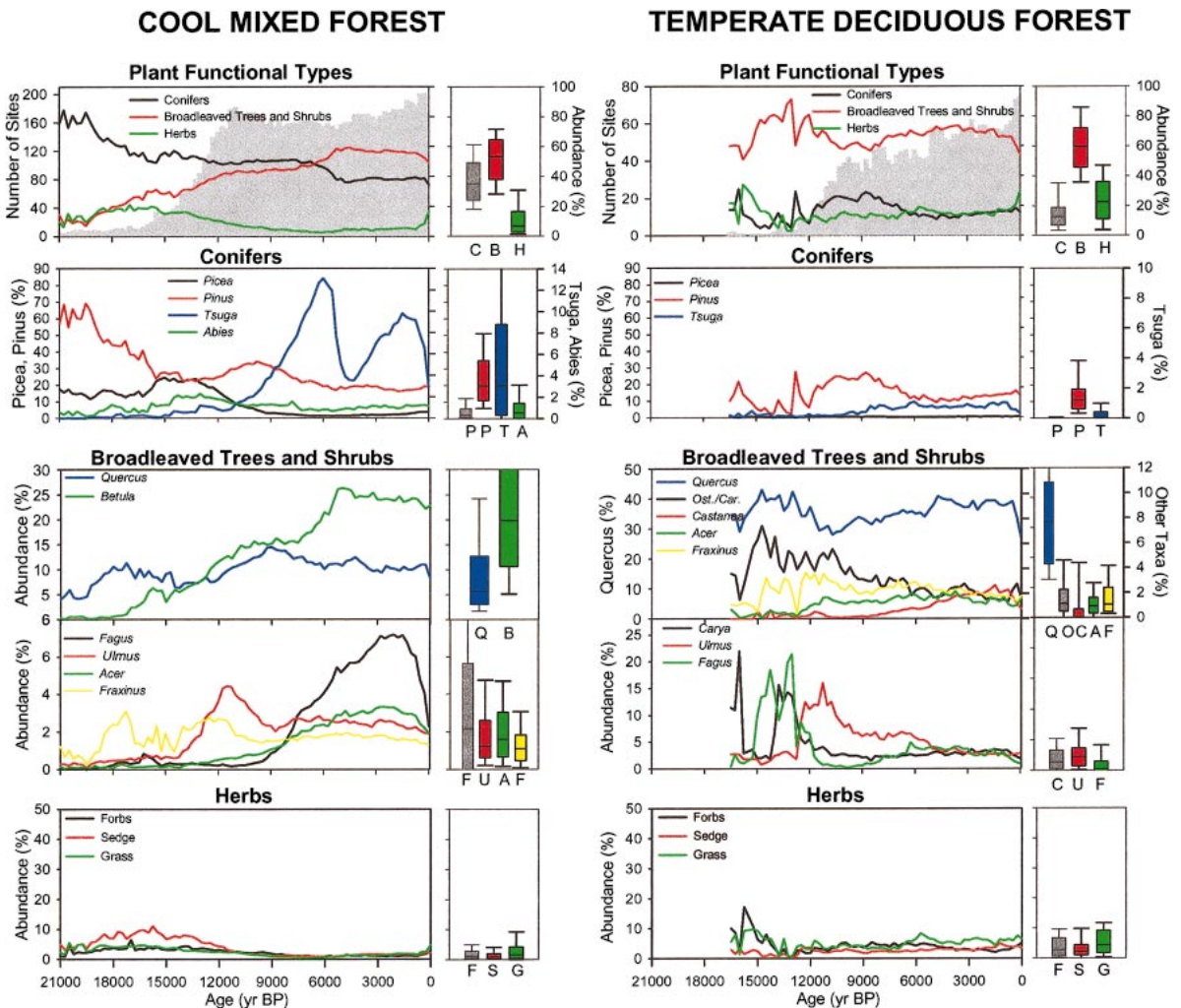


FIG. 6. Plots comparing temporal variation in composition to present-day spatial heterogeneity for the cool mixed forest (CLMX) and temperate deciduous forest (TDEC). The time-series plots show long-term variations of mean pollen percentages for individual taxa and plant life-forms, averaged across all pollen sites assigned to those biomes. Vertical axes represent the percentage pollen abundance for the indicated taxa. The histograms (gray shading) in the top plots show the number of pollen samples (before gridding) assigned to each biome per time interval. Fossil pollen records are distributed nonrandomly in space and time, so the number of samples is only partially related to biome area. At least two pollen records had to be assigned to a biome for an average to be calculated. The present-day spatial variability within each biome is shown in the box plots to the right of each time series. The line bisecting each box is the median, the boxes are bounded by the first and third quartiles, and the whiskers denote the 5% and 95% limits. Note that the right and left y-axes are often scaled differently, but each axis is identically scaled within pairs of time-series and box plots.

panding north in the Canadian Rockies and *Picea* populations expanding from inferred refugia in Alaska (M. E. Edwards, P. M. Anderson, L. B. Brubaker, and A. V. Lozhkin, *unpublished manuscript*). The area of highest abundance for *Picea* pollen has shifted between eastern and western North America several times over the past 21 000 yr. Between 21 000 and 14 000 yr BP peak abundances were located in the interior of eastern North America, but moved into western Canada due to an asymmetrical ice retreat between eastern and western Canada (Ritchie and MacDonald 1986) and did not become centered in eastern Canada until 7000 yr BP, when the near disappearance of the Laurentide Ice

Sheet from eastern Canada altered regional climate patterns and opened new areas for colonization. *Alnus* pollen percentages increased rapidly between 9000 and 7000 yr BP across eastern and western Canada (e.g., Ritchie 1982, Lamb 1984, Anderson 1985; see also footnote 7).

Plant taxa responded individually to past environmental change.—This observation is a central feature of late-Quaternary vegetation history (Cushing 1965, Davis 1976, Webb 1988) and fits well with Gleason's view of plant communities (Gleason 1917, 1926). In a classic example (Davis 1976, Davis 1981b), *Fagus* and *Tsuga* today have closely associated distributions

yet their histories of range and abundance changes differ (Fig. 3a and b). *Tsuga* pollen percentages were below 1% in eastern North America until 14 000 yr BP in the central Appalachians, increased northward along the Appalachian corridor, and by 12 000 yr BP had begun to increase in southern New England. *Fagus* abundances were low until 14 000 yr BP, increased briefly in the southeast between 14 000 and 12 000 yr BP, but did not expand into New England until after 9000 yr BP. The distributions of *Fagus* and *Tsuga* did not attain their modern overlap until the mid-Holocene. Many other taxa that co-occur today—e.g., *Picea* and *Abies*, *Quercus* and *Castanea*, and *Fagus* and *Acer*—have had similar types of differences in history. The temporal changes in taxon distribution and abundance illustrate how plant species respond individually to climate change at continental to regional scales, so that most plant associations have little or no permanence (West 1964).

Individualistic behavior of plant taxa includes the timing and direction of changes in range and abundance, and includes temporal variations in a taxon's location and magnitude of maximal abundance.—As shown for *Fagus*, the range extent of a taxon and its maximal abundance are only partially correlated over time (Figs. 2 and 3). Similarly, shifts in range do not always correspond to shifts in the position of maximal abundance, and plant ranges tend to be more stable than plant abundances. For example, the range of *Quercus* stabilized after 12 000 yr BP, but the region of highest abundances continued to move north from Florida to the central eastern United States, reaching its present position by the mid-Holocene (Fig. 3d). Similarly, the abundance peak for *Picea* shifted more than 1000 km to the east during the Holocene, whereas its range was generally stable after 9000 yr BP, although its abundance increased along its southern boundary during the late-Holocene increases (Schauffler and Jacobson 2002, Fig. 2a). Internal changes in plant population densities at millennial timescales are likely driven by climatically mediated shifts in habitat suitability, seedling and canopy recruitment, adult longevity and competitiveness, population viability, and meta-population size (Webb 1986). Migration is an important mechanism within late-Quaternary vegetation dynamics (Huntley and Webb 1989), but migration alone represents a small part of the full suite of vegetational responses to climate change.

Plant associations

Isopoll maps in which three taxa are mapped simultaneously (Figs. 2b,e and f and 3c and e) provide a level of ecological resolution intermediate between the individual isopoll maps and biome maps. These multi-taxon maps are ideal for showing changes in plant interactions over time (Jacobson et al. 1987). Because each taxon is represented by a single contour, information about plant associations is added at the

expense of less detail about the distribution of individual taxa. The multi-taxon maps represent a bottom-up approach for interpreting the pollen data, because the plant associations are not defined a priori, but rather are chosen based on observed associations among individual taxa.

Plant associations tend to be impermanent features of the landscape.—Individualistic plant behavior scales up to produce ever-changing arrays of plant associations (Figs. 2 and 3). Plant associations have short life spans relative to those of individual plant species (Bartlein and Prentice 1989), and most have not persisted throughout the past 21 000 yr. *Fraxinus* and *Ostrya/Carpinus*, for example, were associated with *Picea* and Cyperaceae between 17 000 and 12 000 yr BP, but are less abundant today and their ranges do not overlap with the abundance peaks of *Picea* and Cyperaceae (Fig. 2e and f). Conversely, the present-day association among *Betula*, *Alnus*, and *Picea* (Fig. 2b), characteristic of the Holocene boreal forest, did not arise until 7000 yr BP. The *Tsuga*–*Fagus* association also began around 7000 yr BP. Plant associations are not random, but neither are they constant and continuous. Of the taxon combinations mapped, rarely do all possible associations occur at once, suggesting that the environmental conditions required to produce all associations are sufficiently different to preclude continuous co-occurrence.

Some taxon associations have persisted for the past 21 000 yr, but some changes in species associations are likely masked by the taxonomic resolution of the pollen data.—*Carya*, for example, has remained closely associated with *Quercus* for the past 21 000 yr (Fig. 3e), although *Carya* abundances have declined since the late Pleistocene (see footnote 7) and *Quercus* abundances increased in New England long before *Carya* (Davis 1976, Prentice et al. 1991). However, both *Quercus* and *Carya* contain several species and it is unknown whether specific associations persisted within the generic *Quercus*–*Carya* association. The late-glacial *Quercus*–*Carya* association in Florida may predominantly represent *Carya floridana* and shrubby oaks (e.g. *Quercus inopina*, *Q. laevis*, *Q. myrtifolia*, *Q. geminata*, *Q. chapmanii*), which grow together in the modern Floridian scrub (Watts 1971; E. C. Grimm, personal communication), which would represent a species association entirely different from that found in the eastern deciduous forests at present. The *Picea*–Cyperaceae association (Fig. 2e) has persisted through time, but moved from the Midwest into the Canadian Cordillera, then northwest into Alaska and east in northeastern Canada by 5000 yr BP. Although the *Picea*–Cyperaceae association has persisted, associations of other taxa with *Picea* and Cyperaceae have changed, e.g., from *Fraxinus* and *Ostrya/Carpinus* during the late Pleistocene to *Betula* and *Alnus* in the Holocene.

Plant range and abundance shifts are less apparent in the multi-taxon isopoll maps.—By focusing on plant

interactions, we see associations emerge, disappear, and move in time and space. However, the directional patterns apparent in the individual isopoll maps are less apparent when the distribution of each pollen taxon is subdivided according to its associations with other taxa. For example, the early-Holocene rise in *Alnus* abundances across Canada, which is strongly apparent in the individual isopoll maps (see footnote 7), is separated in Fig. 2b into the *Alnus*–*Betula* and *Alnus*–*Picea*–*Betula* associations. Plant associations come and go as the distributions of individual plant taxa pass through one another, creating temporary and dynamic associations.

Rates of change

The squared-chord distances between adjacent time periods (Fig. 4) represent a still-higher level of ecological abstraction, in which information about the spatiotemporal distribution of individual taxa or associations is discarded for a general measure of the pacing of vegetational change. The dissimilarity maps alone, however, provide no information about the details of past vegetation change.

Rates of vegetation change have varied both temporally and spatially and are consistent with climatic forcing.—Vegetation development between the last glacial maximum and present can be roughly divided into four stages (Fig. 4): a full-glacial (21 000–17 000 yr BP) stage of relative stasis, transitional stages during the late glacial (16 000–11 500 yr BP) and the early Holocene (11 500–8000 yr BP), and a return to relative stability during the mid- to late Holocene (7000–500 yr BP). Rates of change were fastest during the late Pleistocene and early Holocene (15 000–8000 yr BP), and peak between 13 000 and 10 000 yr BP south of the Laurentide Ice Sheet, coincident with the Younger Dryas Chronozone (Jacobson et al. 1987, Overpeck et al. 1991, Grimm and Jacobson 1992, Shuman et al. 2002b). This event corresponds to the abrupt northward shift of the *Picea*–Cyperaceae association (Fig. 2e), the beginning of the disassociation of peak *Picea* and Cyperaceae pollen percentages (Fig. 2a and c), decreases in *Fraxinus* and *Ostrya*/*Carpinus* abundances (Fig. 2f), increases in *Ulmus* abundances (Fig. 2f), and the rapid expansion westward of *Pinus* (Fig. 2e). High pollen dissimilarities in Beringia between 18 000 and 15 000 yr BP are due to increases in *Betula* pollen abundances and decreases in Cyperaceae, Poaceae, *Salix*, and forbs, marking a switch from a graminoid–*Salix* tundra to one dominated by deciduous shrubs (Anderson 1985, Anderson et al. 1994). In Florida, high rates of change between 17 000 and 16 000 yr BP correspond to decreases in Poaceae and *Artemisia* abundances and increases in *Pinus* and *Carya*.

The effects of European settlement are apparent in the pollen record, and their magnitude is comparable to late-Pleistocene vegetation change.—Between 500 yr BP and present, land-cover changes due to European

land use are signaled by a decline in arboreal pollen taxa, and rises in *Ambrosia*, *Rumex*, and other taxa favored by higher disturbance regimes (Fig. 6; Overpeck et al. 1991, Grimm and Jacobson 1992). The signal is confined to the eastern United States and southern Canada in the region of heaviest population densities and intensive agricultural use (Bernabo and Webb 1977, Foster 1992, Russell et al. 1993, Ramankutty and Foley 1999). *Tsuga* and *Fagus* abundances declined sharply (Fig. 3a and b), as did *Castanea* abundances following the chestnut blight. The magnitude of these declines is comparable to the mid-Holocene *Tsuga* decline (Davis 1981a, Allison et al. 1986). Although the *Tsuga* and *Fagus* decline had begun prior to European arrival, it appears to have been accelerated by increased fire frequency and timbering (Russell et al. 1993). The magnitude of change between 500 yr BP and the present is about one third the net vegetational change between 14 000 and 11 000 yr BP.

Biomes

The biome maps illustrate how species-level plant responses to climate change scale up to alter vegetation physiognomy at continental scales. The cost of abstracting to biomes, however, is loss of information about the distribution and abundance of individual taxa and distribution of plant associations. The biome categories used here are broad (although no broader than typically used), and admit wide compositional variation spatially and temporally. Moreover, like any other vegetation classification, the biomization method must parse continuous variations in abundance into categorical classifications of the vegetation and can lead to misclassifications (Williams et al. 1998). The inferred biomes, therefore, must be checked against the original pollen data to identify any artifacts specific to the classification method.

Biomes are dynamic entities whose distributions emerge from the distributions of individual plant taxa.—The position and area of biomes has changed dramatically over the past 21 000 yr, and most biomes (e.g., mixed parkland, temperate deciduous forest) are not permanent features of the landscape (Fig. 2g). A first-order subdivision of the biome maps separates a late-Pleistocene suite of biomes (parklands, tundra, and cool mixed forest) from 21 000 to 11 000 yr BP and a Holocene suite (tundra, taiga, cool conifer forest, cool mixed forest, temperate deciduous forest, and warm mixed forest) from 10 000 yr BP to the present (Fig. 2g, Webb 1987, 1988). This distribution further subdivides into periods of relative stability during the full-glacial and late Holocene, and rapid transition during the late-glacial and early Holocene (Fig. 2g), consistent with the climatically controlled pacing apparent in the rate-of-change maps (Fig. 4).

Directional trends in distribution shifts are not readily apparent in the biome maps.—Mapping the vegetation as biomes obscures the generally northward ex-

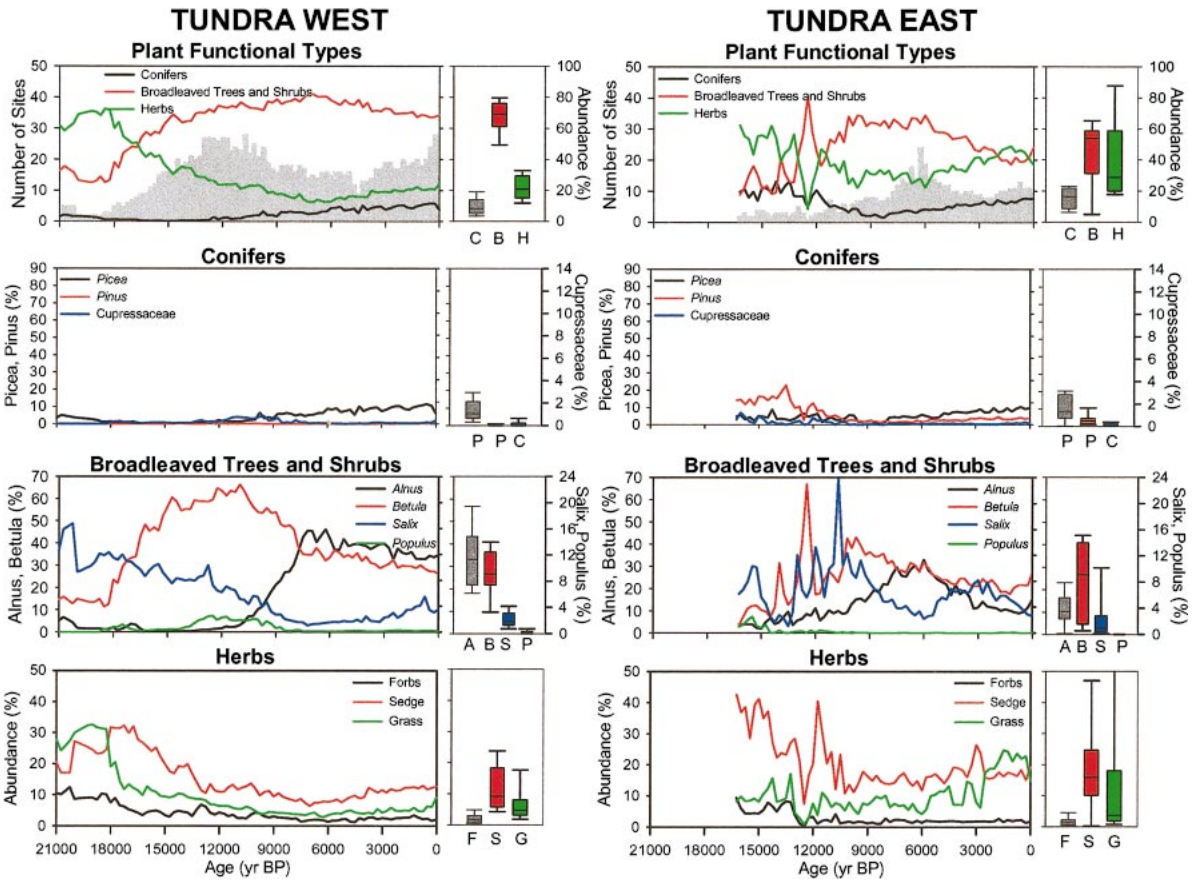


FIG. 7. Plots comparing temporal variation in composition to present-day spatial heterogeneity for the eastern and western varieties of the tundra (TUND) (dividing line set at 105° W). The format is as in Fig. 6.

pansion of plant populations to an even greater degree than the multi-taxon assemblage maps. Individual biomes vary widely in size, emerge and disappear, but do not appear to experience progressive shifts in position when compared to the individual taxa or even the three-taxon assemblages. Two illustrative examples are the mixed parkland, which replaced the western portion of the mixed forest from 16 000–13 000 yr BP, and the taiga, which appeared as the ice sheets retreated after 12 000 yr BP (Fig. 2g). The contrast between the late-Pleistocene and Holocene suites of biomes demonstrates that the first-order feature of the late-Quaternary history of North American biomes is the emergence and disappearance of biomes rather than progressive shifts in distribution.

Biome designations do not always conform to events observed in the maps of individual and associated pollen taxa.—Most of the discrepancies among the biome and isopoll maps can be attributed to the loss of ecological resolution and the transition from continuous data to discrete biome categories. Regional events missed by the biome maps include (1) the development of the *Fagus*–*Tsuga* association in New England between 9000 and 7000 yr BP (Fig. 3c) (occurred within

the cool mixed forest), (2) the replacement of *Picea* and *Cyperaceae* by *Pinus* in the Great Lakes region between 13 000 and 11 000 yr BP (Fig. 2e) (occurred within the cool mixed forest), (3) the rise of *Betula* abundances in Beringia between 18 000 and 14 000 yr BP (see footnote 7) (occurred within the tundra), and (4) the effects of European settlement after 500 yr BP. Changes in the abundance and range of individual taxa with no corresponding change in biome position imply fairly profound changes in biome composition over time.

In a few places, more pronounced discrepancies between the isopoll and biome maps suggest areas where the biome reconstructions do not accurately represent the vegetation. The most significant discrepancy is the underrepresentation of temperate deciduous forest in eastern North America between 11 000 and 9000 yr BP (Fig. 3g). *Quercus* pollen abundances for the same time interval are in excess of 40% (Fig. 3d), levels that have been interpreted to indicate the widespread dominance of *Quercus* and the establishment of deciduous forests (Webb 1988, Overpeck et al. 1992). A second discrepancy is the assignment of cool mixed forest as the predominant forest type in eastern North America for

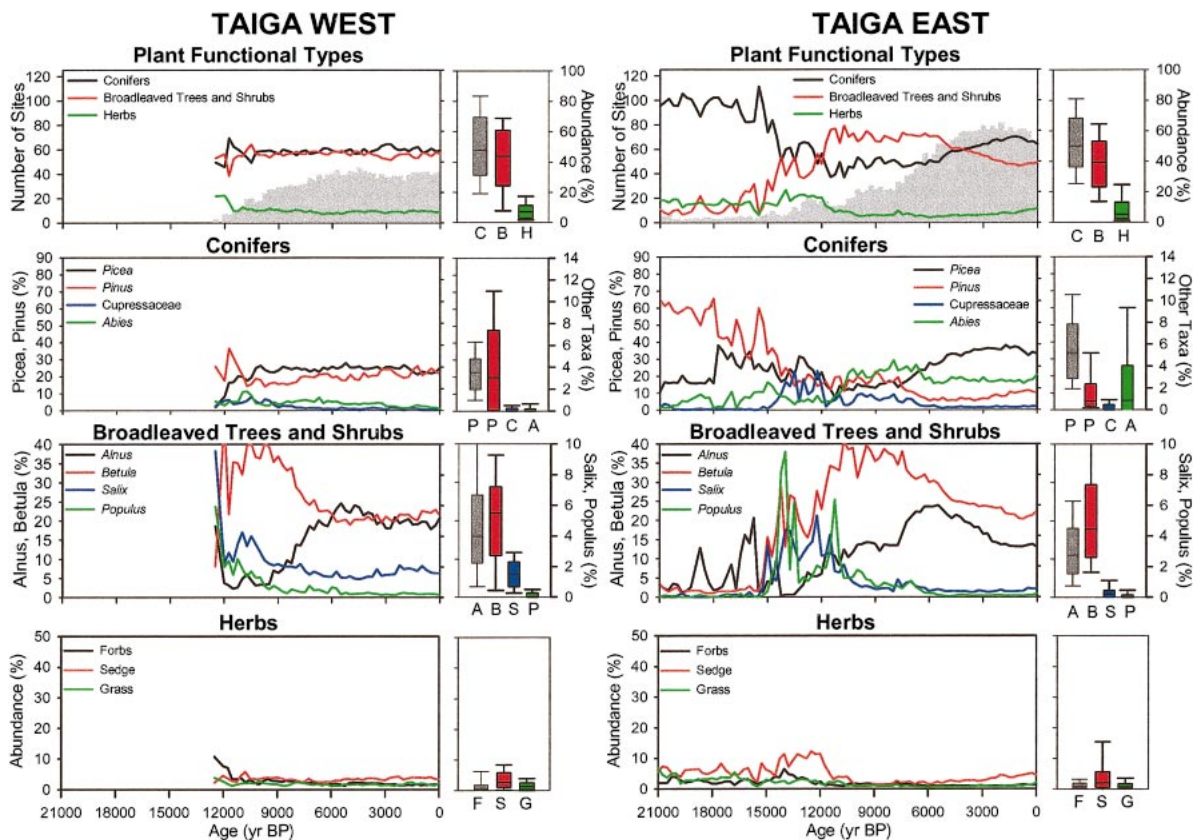


FIG. 8. Plots comparing temporal variation in composition to present-day spatial heterogeneity for the eastern and western varieties of the taiga (TAIG). The format is as in Fig. 6.

the full glacial (Fig. 3g). This assignment is not wholly inaccurate, because the pollen samples contain a mixture of boreal conifer (*Picea*, *Pinus*) and temperate deciduous tree pollen types (*Quercus*, *Fraxinus*, *Ostrya*/*Carpinus*), but it downplays the shift away from the overall dominance of conifers during the full glacial (Fig. 2a and d). The apparent disappearance of tundra from Northern Quebec in the mid-Holocene is an interpolation artifact caused by differences in site densities across the tundra–taiga transition; biome reconstructions for individual sites show that the tundra–taiga boundary was approximately constant between 6000 yr BP and today (Gajewski et al. 1993, Richard 1995, Williams et al. 2000). Finally, the late-Holocene prairie–forest boundary is too far west for similar reasons: an interpolation artifact caused by higher site densities in the eastern forests.

By limiting apparent change to ecotonal zones, biomes underestimate the spatial extent of past vegetational change.—Differences between biome maps occur along biome margins (as a result of the discrete classification), but both the dissimilarity measures and anomaly maps for individual taxa show much broader, continuous changes (Fig. 5). The biome anomalies are qualitatively consistent with the other maps, but the spatial patterns may differ greatly. For example, the

mid-Holocene biome anomalies are consistent with a warmer and drier mid-Holocene in the mid-continent relative to 500 yr BP (Webb et al. 1993), but the anomalies for the biomes are limited to narrow zones: taiga was further north at 6000 yr BP, the prairie–forest boundary was further west, the cool conifer forest was further north, and temperate deciduous forest was further south (perhaps suggesting a regional cooling). The squared-chord dissimilarity (SCD) maps, by contrast, indicate widespread differences between mid-Holocene and pre-settlement vegetation that was not just limited to ecotones. The anomaly maps for individual taxa each span a broader region than the biome anomalies and collectively cover the continent. Although vegetation change can be represented as shifts in ecotones (i.e., borders between biomes), in reality vegetation changes are broad and not limited to ecotonal zones.

The differences between the last glacial maximum and modern vegetation, however, are large enough that the biome anomaly maps are in generally good agreement with the broad-scale changes shown by the other anomaly maps (Fig. 5). The biome anomaly maps show that areas of eastern North America covered by temperate deciduous forest and warm mixed forest at 500 yr BP were cool mixed forest at 21 000 yr BP (Fig. 2g) but that the vegetation of Florida was similar to the

present. This mapping is consistent with the distribution of high SCDs south of the Laurentide Ice Sheet and low dissimilarities in Florida and with the indication that *Picea* and *Pinus* were more abundant in eastern North America (Jackson et al. 2000). In the Pacific Northwest, regions covered by conifer forests at 500 yr BP are mapped as steppe at the last glacial maximum, consistent with the higher-than-present abundances of forb taxa.

Internal variations in biome composition

Changes in the distribution of individual taxa and plant associations that do not correspond to shifts in biome distributions (Figs. 2 and 3) must manifest instead as within-biome variations in composition. These variations are shown in time series plots of the mean pollen percentages for individual taxa and plant life-forms (Figs. 6–11); for comparison, the box plots to the right of each time-series show the present-day variability within each biome. We use the following qualitative descriptions of time-series variation: “small” (mean abundances remain within the interquartile range of present-day spatial variability), “moderate” (the mean abundance remains within the 5% and 95% quantiles), and “large” (the time series variations exceed the 5% and 95% quantiles).

Temporal variations in biome composition over the past 21 000 often exceed the present-day spatial heterogeneity within biomes.—The cool mixed forest, temperate deciduous forest, taiga, tundra, western cool conifer forest, mixed parkland; and warm mixed forest (Figs. 6–11) all display large variations in the relative pollen abundances for one or more plant taxa and life-forms. As a consequence, the biome maps mask substantial differences between full-glacial and modern vegetation. For example, the increase in pollen percentages for *Quercus*, *Alnus*, *Fagus*, and other broad-leaved taxa between the last glacial maximum and present (Figs. 2 and 3) manifests as both an increase in the area of broadleaved and mixed needleleaf-broadleaf biomes (Fig. 2g) and as the proportion of broadleaved taxa within biomes such as the cool mixed forest (Fig. 6) and taiga (Fig. 8). In this way the spatial shifts in taxon distributions translate to compositional shifts within biomes, for example the shift in peak spruce abundances from cool mixed forest to taiga (Figs. 2, 6 and 8). For some biomes, late-glacial swings in composition are likely due to the increased uncertainty caused by low numbers of pollen samples.

Apparent compositional stability for some biomes (e.g., steppe, eastern cool conifer forest) is likely an artifact of the biome definitions and/or taxonomic categories.—Compositionally stable biomes tend to be those that are narrowly defined. The cool conifer forest in eastern North America occupies a narrow zone between the taiga and cool mixed forest (Fig. 2g) and its list of plant functional types closely overlaps with these biomes (Williams et al. 2000). The relatively narrow

definition of the cool conifer forest leaves little room for internal variability (Fig. 9). Therefore, compositional variability is a construct of how broadly the biome types are defined. In the steppe, we have used three broad categories—forbs, grasses, sedges—to represent the herbaceous taxa (Fig. 10), and much of the internal variability in species composition is subsumed within these categories. More detailed analyses of vegetation history in the Great Plains show highly fluctuating abundances of *Ambrosia*, other Asteraceae, Chenopodiaceae/Amaranthaceae, and Poaceae that appear to be linked to seasonal and interannual variations in precipitation (Grimm 2001).

Changes in the frequency of individual taxa frequently scale up to variations in proportions of plant life-forms within biomes.—Among the biomes mapped, the tundra, taiga, cool mixed forest, and warm mixed forest show clear trends in plant life-form composition (Figs. 6–10). Specific trends during the late Pleistocene and Holocene include an increase in the abundances of broad-leaved deciduous trees (*Betula*, *Alnus*, *Quercus*, *Fagus*, *Acer*) in the cool mixed forest (Fig. 6) and taiga (Fig. 8) at the expense of evergreen conifers (*Picea*, *Pinus*), a decline and recovery of evergreen conifers (*Pinus*) in the warm mixed forest (Fig. 10), and, in the tundra (Fig. 7), a long-term decline in herbaceous abundances (Cyperaceae, Poaceae, Forbs) and an increase in broad-leaved shrubs (*Betula*, *Alnus*). These variations in the pollen percentages of plant life-forms likely register long-term variations in biome structure.

DISCUSSION

Different representations of vegetation history

Changes in vegetation over time are multivariate and encompass a wide variety of responses, including changes in a) the range and abundance of individual taxa, b) the associations among taxa, and c) the distribution, structure, and taxonomic composition of biomes. Each representation of the fossil pollen data—whether with pollen diagrams, isopoll maps for single taxa, isopoll maps for multiple taxa, dissimilarity and rate-of-change maps, biome maps, or biome composition plots—highlights different aspects of vegetation history ranging from the behavior of individual taxa to the shifting position and composition of biomes. When taken together, these differing representations of the pollen data provide a holistic view of late-Quaternary vegetation history and reveal a rich variety of vegetation change.

Rate-of-change maps and individual-taxon maps show that the overall pacing of vegetational change (slow between 21 000 and 17 000 yr BP, fast between 16 000 and 8000 yr BP, and slow again after 7000 yr BP) is consistent with climatic control until 500 yr BP, when rates of change re-accelerated due to the effects of European land use (Figs. 2–4). Each plant taxon experienced large and unique shifts in both range and

COOL CONIFER FOREST WEST

COOL CONIFER FOREST EAST

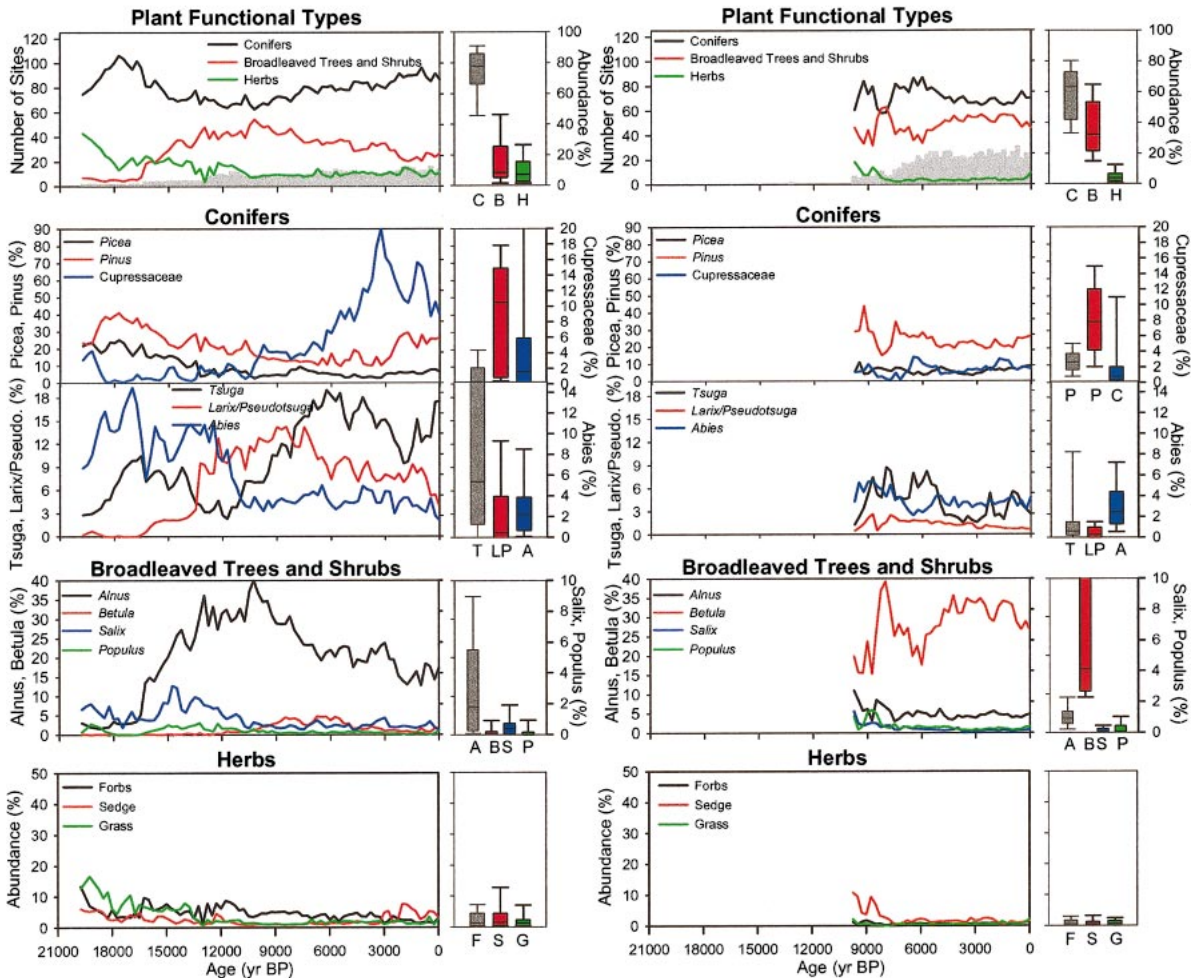


FIG. 9. Plots comparing temporal variation in composition to present-day spatial heterogeneity for the eastern and western cool conifer forests (CCON). The format is as in Fig. 6.

abundance at rates that varied temporally and among taxa (Figs. 2 and 3). Most taxa moved northward (e.g., *Quercus*, *Tsuga*, *Castanea*), but many distribution shifts included east-west components and some spanned eastern and western North America (e.g., *Picea*, *Pinus*) (Figs. 2 and 3). Plant ranges (as approximated by the smallest isopoll contours) tended to stabilize after the early Holocene but abundances continued to vary.

Plant associations have come and gone as a result of the individualistic behavior among taxa (Figs. 2 and 3). Most modern plant associations such as *Fagus*–*Tsuga* and *Picea*–*Alnus*–*Betula* originated during the early Holocene, although a few appear to have persisted throughout the late Quaternary (e.g. *Quercus*–*Carya* or *Picea*–*Cyperaceae*). Other associations appear unique to late-glacial times (e.g., *Picea*–*Fraxinus*–*Ostrya*/*Carpinus*). Biomes, too, are ephemeral at glacial-interglacial timescales, with one suite of biomes characteristic

of the late Pleistocene (cool mixed forest, spruce parkland, mixed parkland, and tundra) and Holocene (tundra, taiga, cool conifer forest, cool mixed forest, temperate deciduous forest, and warm mixed forest). Most biomes have experienced large changes in internal composition over time, with the amount of variation equal to or greater than the spatial heterogeneity within modern biomes (Figs. 6–11). Surprisingly, the mapped shifts in biome position are minor relative to the widespread appearances and disappearances of biomes and large temporal variations in biome composition.

Climate–vegetation interactions

The reciprocal interactions between the vegetation and the atmosphere operate at fundamentally different levels of ecological organization (Fig. 12). At regional to continental and millennial scales, climate is the primary driver of vegetation change, but with many feedbacks active, some significant (Kutzbach et al. 1996,

Broström et al. 1998, Joussaume et al. 1999). Climatic forcing of vegetation operates through direct and indirect controls on the ranges and abundances of individual species. Higher-order properties of the vegetation, such as plant associations and biomes, emerge from these species-level responses, and the atmosphere responds to the altered biogeochemical and biogeophysical fluxes resulting from changes to the emergent property of physiognomy (Bonan 1996, Foley et al. 2000). No single mapping of vegetation properties can therefore represent all aspects of vegetation history, and each highlights different components of vegetation–climate interactions.

Individualistic plant behavior arises in part from intertaxonomic differences in climate tolerance and multivariate changes in climate (Webb 1986, Prentice et al. 1991, Jackson and Overpeck 2000). Dynamic equilibrium exists when the lag of the response variable is short relative to the timescale of the forcing function (Webb 1986). Recent analyses of vegetation responses to the abrupt climate transitions accompanying the Younger Dryas Chronozone and 8200 yr BP event indicate that vegetation response times may be 100 years or less (Ammann et al. 2000, Birks and Ammann 2000, Tinner and Lotter 2001, Shuman et al. 2002a, Williams et al. 2002). Such response times are ample for climate forcings with periods of 1000 or more years (e.g., climate responses to orbital forcing and millennial-scale couplings to ice sheet and ocean dynamics) to produce equilibrium responses in the vegetation. Rates of vegetational change were fastest between 13 000 and 11 000 yr BP (Fig. 4), coincident with the retreat of the Laurentide Ice Sheet between 15 000 and 8000 yr BP (Dyke and Prest 1987) and abrupt atmospheric circulation changes ca. 12 900 and 11 600 ka (GRIP Members 1993, Levesque et al. 1997, Yu and Eicher 1998, Shuman et al. 2002b).

Individualistic plant responses to climate change have caused plant associations to vary continuously and to appear and disappear (Figs. 2 and 3). Each plant taxon has a unique fundamental niche in n -dimensional environmental space (Hutchinson 1957) that includes climatically controlled plant population densities within the niche. Plant associations occur when the realized environmental space overlaps with the fundamental niches for a set of taxa (Jackson and Overpeck 2000). The amount of overlap among fundamental niches influences the likelihood of species association, but the non-overlap in niches guarantees different responses as the environment changes. Plant responses to climate change therefore are individualistic, but usually not entirely independent due to cross-correlations among plant niches. The persistence of plant associations such as *Quercus*–*Carya* (Fig. 3e) suggests that those taxa have highly coincident fundamental niches, whereas more transient plant associations (e.g., *Fraxinus*–*Picea*, *Fagus*–*Tsuga*) indicate less overlapping among fundamental niches. The apparent size of the funda-

mental niche also varies with the taxonomic resolution available from the pollen data, so that generic associations such as *Carya*–*Quercus* may have more permanence than specific associations (e.g., *Fagus grandifolia*–*Tsuga canadensis*). More-transient associations depend on particular combinations of environmental conditions, such as winter minimum temperatures and growing season strength and length, which change over time due to changes in the seasonality of insolation and the atmospheric concentration of greenhouse gases. Past plant associations with no modern analog likely grew in environments outside the range of modern climate space (Prentice et al. 1991, Williams et al. 2001).

Shifts in species distributions and densities also scale up to alter vegetation physiognomy and structure, represented in our maps as changes in biome type. Because biomes are defined as assemblages of plant functional types, movements of biomes do not imply that plant communities moved as whole floristic units—a classic Clementsian view that is inconsistent with Quaternary evidence (Davis 1976, 1981b). Rather, the distribution of plant functional types and biomes emerges from the movements of individual species, and the possible combinations of plant functional types (i.e. biomes) are limited relative to the number of species combinations.

The broad changes in vegetation structure represented by the biomes likely affected carbon sequestration and the terrestrial carbon cycle (Crowley 1995, Peng et al. 1998), aerosol source areas (Harrison et al. 2001, Kohfeld and Harrison 2001), and other interactions between the vegetation and atmosphere (Kutzbach et al. 1996, Broström et al. 1998, Joussaume et al. 1999). Not all abundance changes for species and plant functional type, however, correspond to shifts in biome distribution, but instead may often be manifested as internal variations in biome composition (Figs. 6–11). Informed use of the pollen-based biome reconstructions therefore requires an appreciation of which aspects of vegetation change the biome maps capture, and which are missed.

Assessing the biome reconstructions

The biome reconstructions provide a first-order description of the past vegetation that is broadly consistent with the plant taxon distributions apparent in the individual isopoll maps. Analog-based methods have shown that a large area of vegetation ca. 14 000 yr BP (12 000 radiocarbon years BP) had no modern analog (Overpeck et al. 1992). The biomization method classified this vegetation as “mixed parkland,” a biome absent in North America at present (Williams et al. 2001). The timing of the reorganization of biomes between the Pleistocene and Holocene is consistent with the rate-of-change maps (Fig. 4), and the emergence of individual biomes can be traced to the emergence of new associations among taxa (Figs. 2 and 3). For example, the development of taiga in western Canada follows the northwestward expansion of *Picea* and the

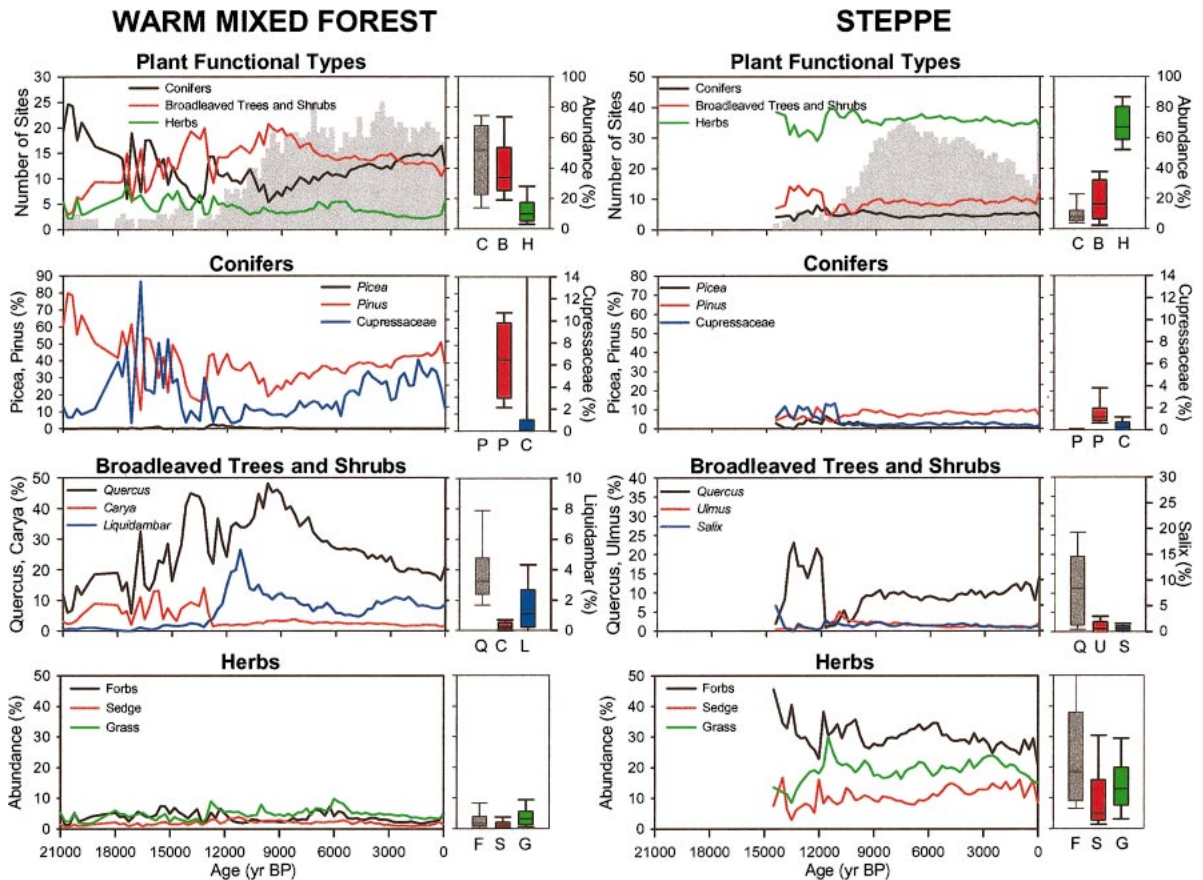


FIG. 10. Plots comparing temporal variation in composition to present-day spatial heterogeneity for the warm mixed forest (WMMX) and steppe (STEP). The format is as in Fig. 6.

subsequent increase of *Pinus*, *Betula*, and *Alnus* abundances (Fig. 2a,b, and d).

Biomes have limitations inherent to any categorical classification of the vegetation (DeFries et al. 1999). First, they are subject to classification error. Second, they cannot fully represent the continuous nature of vegetation gradients in time and space, and they limit apparent vegetation change to ecotonal boundaries (Fig. 5). Third, information is lost in scaling up from individual taxa to biomes, information that is relevant to paleoclimatic inference. Fourth, the biomes mask large within-biome variations in plant abundances and vegetation structure (Figs. 6–11), thereby hiding differences between late-Quaternary biomes and their modern counterparts. The cumulative effect is to downplay the overall magnitude and variety of vegetation change during the late Quaternary.

Discrepancies among the isopoll and biome maps are due in part to the difficulty of imposing categorical classifications upon continuous multivariate entities such as vegetation, so areas where the classifications do not fit cleanly are to be expected. However, some discrepancies can be attributed to the biomization method. Misclassifications are most common among

biomes with similar floristic lists. For such biomes, the presence or absence of key taxa is crucial, making the biomization technique highly sensitive to pollen abundances near the threshold limit (Williams et al. 2000). For this reason, the gridded biome maps for the last glacial maximum (Fig. 2g) are similar but not identical to the biome assignments for individual pollen sites (Williams et al. 2000, Fig. 3). Early-Holocene pollen records misassigned to cool mixed forest instead of temperate deciduous forest typically have high abundances of *Quercus* (>40%) and low abundances of *Picea* and *Tsuga* (>1%) (Figs. 2 and 3). When selecting between the cool mixed forest and temperate deciduous forest, the biomization technique is not sensitive to high or low *Quercus* percentages, but is sensitive to the presence of *Picea* and *Tsuga*. In general, taxonomically diverse pollen samples tend to be assigned to cool mixed forest because this biome includes the most taxa (Williams et al. 2000) and so tends to run up the highest affinity scores. The insensitivity of the biomization method to variations in abundance is relevant because the physical properties of the vegetation are influenced both by the mixture of species present and the relative abundances of each species.

Biome anomaly maps limit apparent vegetation change to ecotonal boundaries and thus imply vegetational stability in interior regions of biomes (Fig. 5). This perspective, however, is inconsistent with the anomaly maps for many taxa (Fig. 5), which show that vegetation change was spatially widespread (Bernabo and Webb 1977) and involved compositional changes within biomes (Figs. 6–11). The biome and taxon maps qualitatively agree—for example, biome maps for the mid-Holocene show an expanded steppe and cool conifer forest relative to the present, consistent with elevated abundances of prairie forbs and *Pinus* pollen—but biome anomalies capture only a fraction of the area experiencing large changes in vegetation composition (Fig. 5). The biome maps therefore contribute to the perception that vegetation sensitivity to climate change is greatest at ecotones (e.g., Peteet 2000, Yu 2000), a perspective not borne out by more finely-resolved representations of the pollen data. Because changes in vegetation and climate may be hidden by the lack of change in biome maps, data–model comparisons based upon comparing climates to biome shifts may be misleading (CAPE Project members 2001).

Because vegetation responses to climate change manifest at the scale of individual taxa (Fig. 12), classifying the pollen data to biomes inevitably downplays the full scope of vegetational responses to past climate change and so misses key climatic signals (Gajewski 1993). For example, the rapid expansion of *Pinus* in Ohio and Indiana during the Younger Dryas Chronozone (Fig. 2), was likely caused by a regional decline in moisture (Shuman et al. 2002b). Similarly, the Holocene increase of *Tsuga* and *Fagus* in the northeastern United States and southern Canada (Fig. 3) has been linked to increases in summer temperatures and precipitation after the collapse of the Laurentide Ice Sheet (Prentice et al. 1991, Shuman et al. 2002b). Neither of these vegetational responses to shifts in moisture is apparent in the biome maps (Fig. 2).

Changes in the distribution of individual taxa and plant associations that do not correspond to changes in biome position or the appearance or disappearance of biomes (Figs. 2 and 3) must be manifested instead as within-biome shifts in the relative abundance of pollen taxa (Figs. 6–11). The temporal variations in mean biome composition generally match or exceed present-day spatial heterogeneity within each biome. Changes in the proportions of individual plant taxa often scale up to affect the relative proportion of plant functional types (Figs. 6–11). Changes in the proportions of plant life-forms within a biome are possible in the biomization method because its biome definitions only specify a list of functional types, not relative proportions. Therefore, a pollen assemblage with 1% *Picea* (a boreal evergreen conifer) and 99% *Betula* (a boreal deciduous tree) and another with 99% *Picea* and 1% *Betula* would be equally valid examples of the taiga under this methodology, but would have contrasting composition and

physical properties and represent different plant communities. Consequently, large geographic shifts in the peak abundance of many taxa, which constitute a significant change in taxon distributions, are not always captured by the biome maps.

Physical properties of the land surface are often estimated based upon biome type, but changes in vegetation structure within biomes weaken the assumption that the modern physical properties of biomes may be assigned to their past counterparts. In general, the proportion of pollen types from herbaceous taxa within forested biomes has decreased over time, indicating a trend from more-open to more-closed forests, and the proportion of pollen types from broadleaved deciduous taxa has increased over time (Figs. 6–11). For example, the full-glacial cool mixed forests and taiga apparently had higher abundances of evergreen conifers relative to their Holocene counterparts. The higher abundance of conifers should have reduced the seasonal fluxes of carbon, water, and radiation between the vegetation and atmosphere (Bonan 1995, Sellers et al. 1997). In the tundra, the observed increase in deciduous shrub abundances between 18 000 and 13 000 yr BP may have increased seasonal variations in albedo and partitioning between sensible and latent heat fluxes (Chapin et al. 2000). Attributing carbon densities from modern biomes to their full-glacial counterparts will produce carbon sequestration estimates that are too high if full-glacial forests were more open than their modern counterparts (Adams et al. 1990, Prentice and Fung 1990, Prentice et al. 1993, van Campo et al. 1993, Crowley 1995, Adams and Faure 1998).

Vegetation reconstructions are needed that accurately describe the physiognomic properties of vegetation in a continuous fashion, thereby avoiding the limitations inherent to categorical classifications. Reporting the affinity scores for each biome calculated by the biomization method, rather than the final biome assignments, provides a continuous index of vegetation change (Marchant et al. 2002), but the precise relationship between affinity scores and vegetation physiognomy is unclear. Another approach is to aggregate the pollen percentages by plant functional type (Peyron et al. 1998, Peyron et al. 2000), although this aggregation may be hampered by intertaxonomic differences in pollen productivity (Williams and Jackson 2003). Reconstructions of late-Quaternary tree cover (Williams 2003), based upon the application of modern-analog methods to remotely sensed vegetation data sets (DeFries et al. 1999; 2000), provide continuous descriptions of fractional cover for broadleaved, needleleaved, and herbaceous plant functional types. These tree-cover maps provide a new kind of benchmark for data-model comparisons for coupled vegetation–atmosphere models (Foley et al. 1998, 2000, Ganopolski et al. 1998a, b) as well as a perspective into late-Quaternary vegetation history complementary to those presented in this paper.

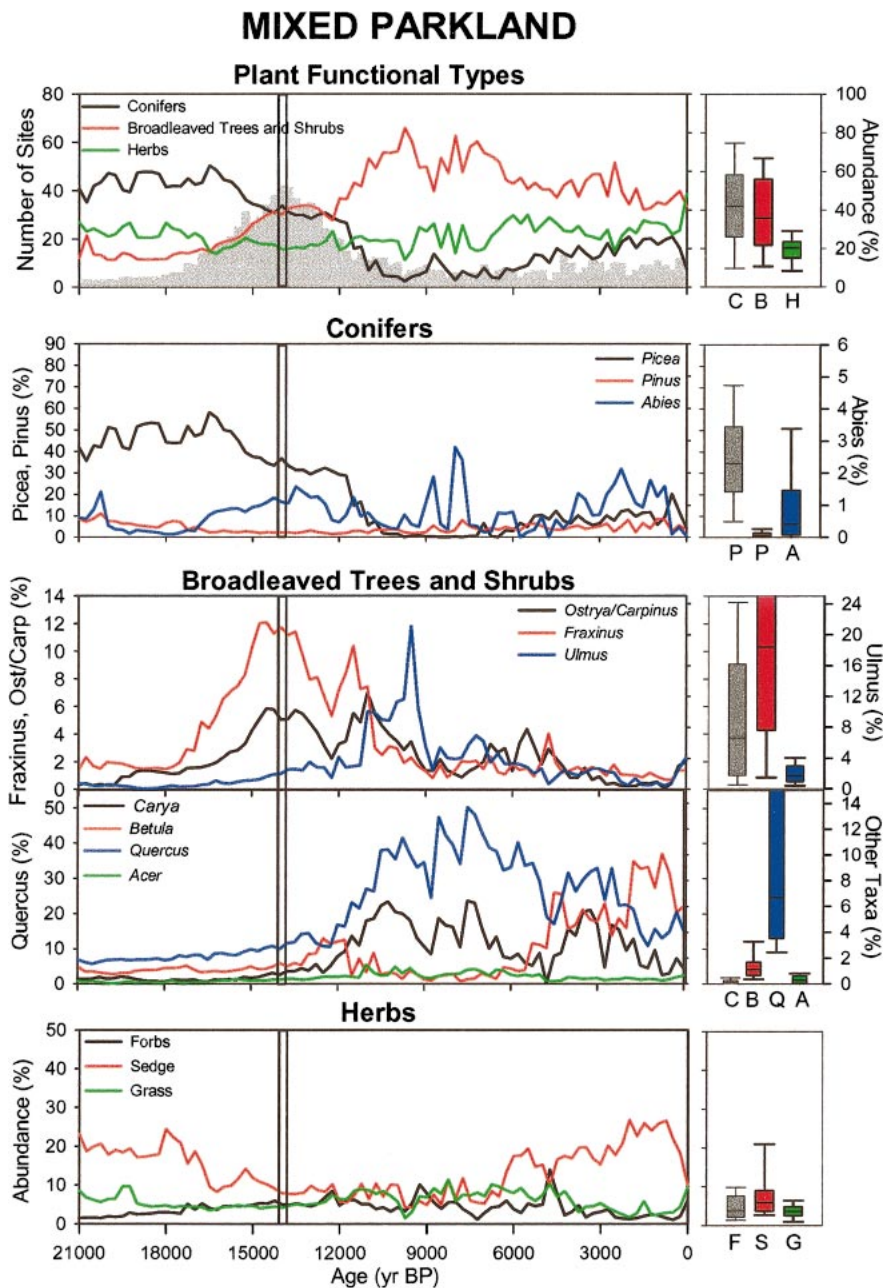


FIG. 11. Plots comparing temporal variation in composition to present-day spatial heterogeneity for the mixed parkland (MXPA). The format is as in Fig. 6 except, because the mixed parkland is nearly extinct today, the box plots show the spatial variability for 14 000 yr BP (marked by vertical bar in time series).

CONCLUSIONS

Vegetation responses to late-Quaternary environmental change have been rich and varied, and have been expressed as variations in (1) the range and abundance of individual taxa, (2) the associations among plant taxa, (3) the temporal and spatial patterns of rates of change, and (4) the distribution, composition, and structure of biomes. Climatic control of vegetation change at regional to continental and millennial

scales is exerted at the level of individual taxa, from which higher-order properties of the vegetation emerge. The pacing of vegetational change was slow between 21 000 and 17 000 yr BP, accelerated during the late-glacial (16 000–11 500 yr BP) and early Holocene (11 500–11 800 yr BP), and slow between 7000 and 500 yr BP. *Plant taxa* have experienced individualistic (but not entirely independent) shifts in range and abundance, and responses to climate change were

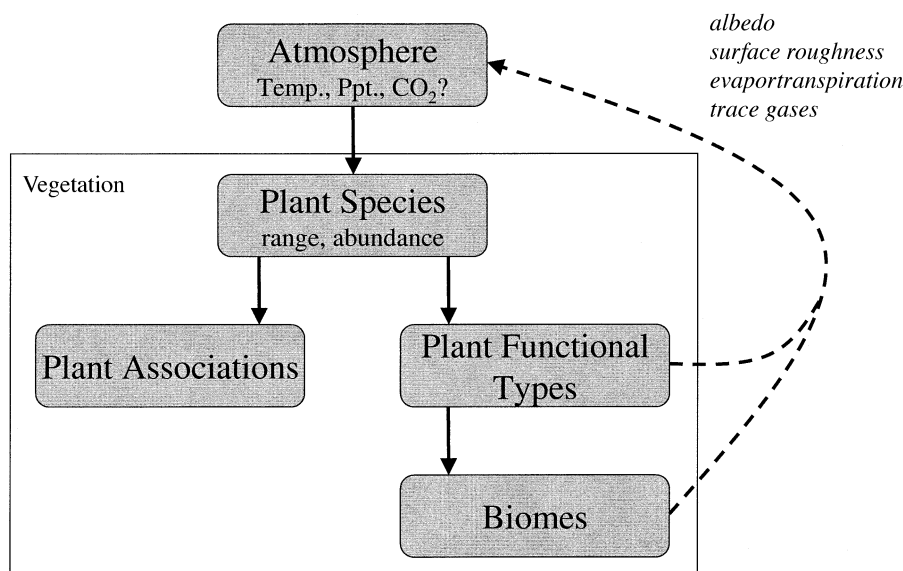


FIG. 12. Schematic diagram of the reciprocal interactions between the vegetation and the atmosphere. Species distributions at regional to continental scales are primarily determined by climate; variation in climate is the primary driver of vegetation dynamics at millennial timescales. Individualistic species responses result in continuously changing associations among plants and long-term variations in vegetation physiognomy, described as the distribution of plant functional types and biomes. Gross changes in vegetation structure alter the physical properties of the land surface, modulating the exchanges of energy, moisture, and carbon between the atmosphere and vegetation.

not limited to range margins. *Plant associations* develop as the distributions of individual taxa intersect one another, with new associations appearing and disappearing over time. *Biomes* also emerge and vanish over time, and vary in composition temporally and spatially, but generally move less than the spatial redistributions apparent for individual taxa or even the more muted movements of associations. These changes in biome distribution and structure imply fundamental shifts in vegetation structure in North America that in turn have altered the biogeophysical and biogeochemical fluxes between the land surface and atmosphere. Each kind of representation of the vegetation provides a unique perspective into late-Quaternary vegetation history, and together allow a fuller understanding of the ecologically asymmetrical scaled interactions between the vegetation and the atmosphere. Biome reconstructions are useful for describing past land cover at a scale suitable for interfacing with Earth-system models, but they carry significant limitations inherent to categorical representations of the vegetation. The biome maps provide a useful first-order characterization of the land surface but downplay the overall variety and magnitude of vegetation change and miss climatically significant events apparent in the individual-taxon maps. In some places, the biome assignments may be in error, particularly between biomes with similar floristic lists. The biome maps focus apparent vegetation change at ecotones, whereas actual changes were more widespread. The observed temporal variations in biome composition

are analogous to the spatial heterogeneity within modern biomes, but are of equal or larger magnitude. Compositional changes at the level of individual plant taxa can scale up to affect the relative abundance of plant functional types within biomes and thus weaken the assumption that the biogeophysical and biogeochemical properties of modern biomes can be applied to their ancient counterparts. Biome maps, therefore, must be used with caution, for they imply an internal homogeneity that does not exist. Most of these limitations are inherent to categorical classifications of the vegetation and can be avoided by developing continuous indices of vegetation structure.

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SUPPLEMENT

GIF-animated version of pollen-percentage maps and gridded pollen data sets used for mapping are available as a Supplement in ESA's Electronic Data Archive: *Ecological Archives* M074-007-S1.